Position Control in Lithographic Equipment [Applications of Control]

Article *in* IEEE Control Systems Magazine · November 2011

DOI: 10.1109/MCS.2011.941882 · Source: IEEE Xplore

Hans Butler
ASML

Position Control in Lithographic Equipment

An Enabler for Current-Day Chip Manufacturing

HANS BUTLER

Lithographic steppers and scanners are highly complex machines used to manufacture integrated circuits (ICs). These devices use an optical system to form an image of a pattern on a quartz plate, called the reticle, onto a photosensitive layer on a substrate, called the wafer. The circular wafer, having a diameter of 200 or 300 mm, is usually made of silicon. Since one wafer can contain many ICs, typically 100 or more, the wafer needs to be repositioned from exposure to exposure. Exposure itself takes place during a scanning motion of the wafer and the reticle.

The chip manufacturing industry continually strives to pack more functionality into each IC. This progress has had a major impact on the capacity of memory chips and the operating speeds of microprocessors. Moore's law, which states that every one and a half to two years the computing power per chip doubles, has held steady for four decades [1]. Figure 1 shows an example of the result for consumers in the form of the price per gigabyte of NAND flash memory, which amounts to a thousand-fold decrease in slightly over ten years. The minimum feature size, indicative of the smallest component that can be manufactured in one IC, is limited by the wavelength of the imaging light, the refractive index of lens materials, and ultimately the size of individual atoms of the substrate material. Today, this minimum feature size is about 20 nm [2], with 1 nm being a millionth of a millimeter.

Three elements make lithographic tools challenging from a position-control perspective. First, the lithographic tool needs to be accurate to support the small feature sizes of the IC patterns that are being exposed. During exposure, the wafer and reticle are moved with constant velocity. The allowable position error during the scan is only a few nanometers, a fraction of the minimum feature size. Measurement systems, environmental conditioning, and control tactics are critical elements in obtaining such a high accuracy.

Second, these machines need to process a large number of wafers per hour. A throughput of 175 wafers per hour is not uncommon, allowing a processing time per wafer of only 20 s, or 0.2 s per exposure. A wafer scan speed during exposure of 0.6–1 m/s is standard. The force required for the high acceleration of 20–40 m/s^2 can induce vibrations, which conflict with a nanometer-scale positioning accuracy. Reticle motion is typically a factor of four faster, with correspondingly higher accelerations.

Finally, because these machines are used in an industrial environment, robustness and reliability are essential. ICs with smaller features, which provide a higher profit for the chip manufacturer, require the latest technology in the photolithographic scanner. Bringing the latest technology to market early tends to conflict with the reliability requirement, since this technology typically takes a considerable amount of time to optimize due to its complexity. In addition, because of production numbers, the lithographic tool's design needs to be robust, avoiding elaborate manual adjustments on a machine-to-machine basis.

Table 1 summarizes progress in stepper and scanner technology between 1994 and 2011. By doubling the wafer diameter, tripling the number of wafers processed per hour, and reducing the smallest feature size by a factor of 17.5, the number of bits per hour produced by a lithographic tool has increased by a factor of 2200.

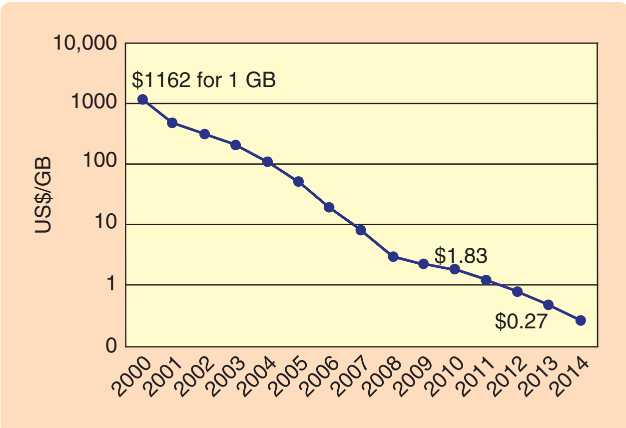

FIGURE 1 Memory price over the years (in U.S. dollars). One gigabyte of NAND flash memory, costing more than US$1000 at the beginning of this millennium, has now dropped in price to about US$1. A further price decrease is anticipated. This price drop enables the usage of flash memory in portable equipment. Future improvements depend on the ability to create more memory bits per square centimeter of wafer surface, keeping production cost steady while increasing memory capacity. (Source: Gartner, January 2011.)

Digital Object Identifier 10.1109/MCS.2011.941882
Date of publication: 16 September 2011

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The high speeds and accelerations needed for wafer and reticle motion are in most cases produced by linear motors that exert reaction forces on the support structure. These forces can be as large as 5000–10,000 N. The dynamic design of the machine must ensure isolation of these reaction forces from the position measurement systems and the projection optics. Remaining vibrations in these components must be damped and what remains needs to be taken care of by the stage control systems. After the acceleration phase, a few milliseconds of settling time are allowed for the stage to reach nanometer scanning accuracy.

This article describes the basic optical principles in the lithographic tool, with the resulting positioning accuracy requirements. For three generations of lithographic tools, the mechatronic architecture and control implications are discussed. Then, six degrees of freedom (DOF) stage control is described with the main focus on actuator force decoupling, allowing the use of classical single-input, single-output (SISO) controllers.

THE LITHOGRAPHIC PROCESS IN SEMICONDUCTOR MANUFACTURING

In semiconductor manufacturing, the wafer is covered with a layer of photosensitive ink called resist. The optical system in a lithographic stepper or scanner exposes a reticle pattern onto the wafer, creating an image in the resist. The image size on the reticle is four times as large as the projected image on the wafer, since the projection lens reduces the image size by a factor of four. The maximum size of one exposure area on the wafer, called a die, is typically 26×32 mm; one die can hold multiple separate IC images. Figure 2 shows a basic picture of the lithographic process.

A source produces light that is captured by the first part of an optical system, called the illuminator. The illuminator modifies the optical properties of the light beam, such as polarization and pupil shape, and transmits the light toward the reticle. Monochromatic light is required to avoid optical aberrations, caused by the wavelength dependence of the refractive index of lens materials. After passing through the reticle the light enters the projection lens, which creates an image on the wafer.

To expose a die, the wafer is moved to bring the die under the lens. The light source is then switched on until the amount of light radiation on the wafer reaches the required dose. The wafer is then positioned for the next exposure and the cycle repeats itself. When all of the dies on the wafer are exposed, the wafer is removed from the lithographic tool and is replaced by a new one. The exposed wafer is then subjected to several chemical, mechanical, and thermal processing steps. After finishing these processing steps, the wafer is again covered by a photosensitive layer and brought into a lithographic tool. This tool is not necessarily the same one used in exposing the previous layer. Another pattern is then exposed exactly on top of the previous one, and, after exposing the complete wafer, the wafer is again removed from the lithographic tool to undergo new process steps. The complete cycle repeats itself up to 30–60 times until the stack of layers on the wafer make up a working electronic device. All lithographic tools that a wafer encounters must behave similarly to avoid differences between functional layers of the chip. Machine-to-machine calibration, called matching, is therefore performed.

In contrast with the lens schematically depicted in Figure 2, a real lens consists of 10–30 optical elements. The wafer is placed on top of a flat table, the wafer stage. Illumination wavelengths λ of 365, 248, 193, 157, and, recently, 13.5 nm, are being used [3], depending on the applied source type. Since the exposure process is diffraction limited, a smaller wavelength enables creation of a smaller pattern on the wafer. Smaller structures enable higher density of the

TABLE 1 Requirements on lithographic equipment in 1994 and 2011. A short comparison between lithographic tool properties in 1994 and 2011, indicating a 2200-fold increase in the number of manufactured bits per hour. Note that this number scales quadratically with the minimum feature size. A 300-mm wafer can hold 2.5 times as many chips as a 200-mm diameter wafer, which is more than the wafer surface increase due to the relatively smaller nonusable area at the wafer edge.

	1994	2011
Wafer diameter [mm]	200	300
Productivity (wafers per hour)	60	175
Minimum feature size [nm]	350	20

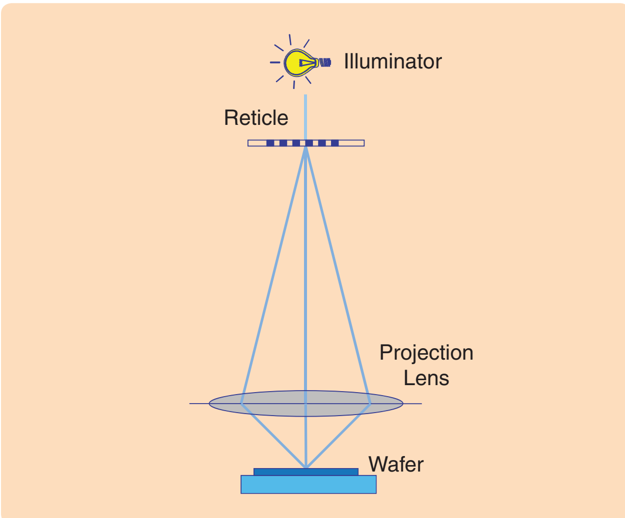

FIGURE 2 The lithographic process. Light, usually produced by a laser source, passes a quartz reticle holding a chrome pattern of one chip layer. The diffracted light passes the projection lens that forms an image on the wafer, which is held by a table. The reticle and the wafer are both movable by a positioning system.

electronic circuit in the chip to be made, enabling a higher CPU processing capability and memory size.

Basic Imaging

The projection light's wavelength λ is of the same order of magnitude as the pattern pitch on the reticle, and thus diffraction effects must be taken into account. Assume that the reticle pattern consists of evenly spaced lines with a spatial period L, which is true especially for memory chips. The reticle behaves as an optical grating and diffracts the incoming beam into angles α_n given by

$$\sin(\alpha_n) = \frac{n\lambda}{L}, \tag{1}$$

where the integer n denotes the *order* of the exiting light path. Not all orders have the same intensity. The 0th order, which exits the reticle with the same angle as the incoming light, has an intensity I_0 that equals half the intensity of the incident light on the reticle. This decrease in intensity is caused by the lines on the reticle, which block half the incoming light. The intensity of the even orders equals zero; the intensity of the odd orders equals $(2/\pi n)I_0$, as further illustrated in Figure 3.

The diffraction orders can be considered to be components of a Fourier series. All components together make up the intensity pattern

$$I(x) = I_0 + \frac{2}{\pi} I_0 \sum_{k=-\infty}^{\infty} \frac{1}{(2k+1)} \sin\left(\frac{2\pi(2k+1)x}{L}\right) \tag{2}$$

at the reticle level, where L is the pattern pitch and x is the position in the pattern, as indicated in Figure 3. Not all diffraction orders at the reticle level can be captured by the projection lens and imaged onto the wafer. In particular, only orders that have a diffraction angle $\alpha_n < 90^\circ$ can be captured and imaged. The block-like pattern on the reticle does not reproduce itself as a block-like pattern on the wafer since the higher Fourier orders are cut off. If only the 0th and $\pm$ 1st order are captured by the lens, a sinusoidal pattern on the wafer surface remains since only the 0th and $\pm$ 1st components in the Fourier series recombine. The photosensitive resist typically reacts suddenly after a certain threshold in dose has been reached and transforms the sinusoidal image into a block-like structure in resist. Imaging up to the $\pm$ 1st orders is therefore sufficient to recreate lines and spaces on the wafer. If only the 0th order is captured by the lens and projected onto the wafer, then only an average intensity I_0 on the wafer remains and imaging of the pattern no longer takes place. The sine of the maximum angle that a lens is able to image onto the wafer is called the numerical aperture (NA). The practical limit lies around NA = 0.93. Adding a layer of water between the projection lens and the wafer, called immersion lithography [4], increases the NA by the index of refraction of water, which equals 1.44 at the popular 193-nm wavelength. The minimum corresponding pitch L of the imaged pattern on the wafer is given by (1) with $n = 1$ and $\sin(\alpha_1) = \mathrm{NA}$

$$L = \frac{\lambda}{\mathrm{NA}}. \tag{3}$$

In a current-day immersion tool, with $\lambda = 193$ nm and $\mathrm{NA} = (0.93)(1.44)$, the minimum pitch L thus equals 144 nm. By modifications in the optical system, the minimum pitch can be halved to 72 nm. In the IC manufacturing industry, the half pitch is usually mentioned, being 36 nm. This value is the minimum theoretical *critical dimension* in a one-step exposure process. By using multiple-exposure techniques, smaller elements can be manufactured [5].

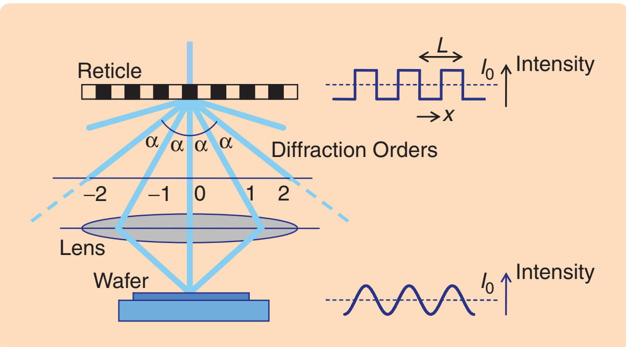

FIGURE 3 The basic imaging process. Light is diffracted into orders by the fine patterns on the reticle. The projection lens captures only the lower orders of the diffracted light. A repeating line-space pattern on the reticle is then imaged as a sinusoidal intensity pattern on the wafer. The characteristics of the photosensitive layer ensure that, after development and etching, a block-like resist pattern on the wafer remains.

Positioning Accuracy Specifications

Figure 2 shows the reticle and the wafer fixed with respect to the projection lens. In practice, the reticle and wafer are placed onto positioning tables, which are actively controlled. A positioning error of either of the two tables results in shifting the image projected on the wafer in the horizontal direction or results in bringing the wafer surface away from the lens focal plane. Two specifications on the lithographic tool determine the required accuracy of both stages.

First, *overlay* is the ability of the lithographic tool to expose two images exactly on top of each other. A functional chip requires all chip layers to cooperate with each other. Overlay is directly influenced by stage-positioning accuracy, in addition to numerous other contributors. The average position error of the stage during exposure determines the location where the image is placed on the wafer. This measure is called *moving average* (MA), which is defined by

$$\mathrm{MA} = \frac{1}{T}\int_{-T/2}^{T/2} e(t)\,\mathrm{d}t. \tag{4}$$

In (4), T represents the exposure time, and $e(t)$ represents the position error as a function of time t.

Second, the imaging process is degraded by a limited stage-positioning accuracy since position errors reduce contrast, similar to taking a photograph while not keeping the camera still. Here, the *moving standard deviation* (MSD) of the positioning error during exposure is the critical measure, with

$$\mathrm{MSD} = \sqrt{\frac{1}{T}\int_{-T/2}^{T/2} (e(t) - \mathrm{MA})^2 \mathrm{d}t}. \qquad (5)$$

MSD represents the higher-frequency part of the positioning error, while MA represents the lower-frequency part of the error, the boundary being determined by the exposure time T. For 38-nm half-pitch lithography, MA must be limited to about 1 nm, while MSD must be limited to about 7 nm. The exact limit depends on the manufacturing process characteristics such as resist properties, repeating elements in the image pattern, and the ratio between line size and space size.

Focus errors in the form of MA and MSD of the positioning error in the z direction, parallel to the optical axis of the lens, also contribute to imaging errors. The focus error can typically be allowed to be 10× larger than the horizontal positioning error.

THE STEPPER

Basic Operation

The stepper, schematically shown in Figure 4, was the main lithographic tool used until about 1997. Each die on the wafer is exposed in a flash, after which the wafer is repositioned to the next die location. This process repeats itself until the full wafer is exposed.

The projection lens is mounted onto a metrology frame, called a metroframe, which is supported by pneumatic vibration isolators connected to the base frame, which is attached to the floor. The isolators, functioning as weak springs with a decoupling frequency in the 1-Hz range, block floor vibrations from entering the metroframe.

The reticle is positioned on a table attached to the metroframe. A robot system provides automatic reticle exchange. Also connected to the metroframe is the illuminator, which provides the illumination light. The illuminator-to-metroframe connection is possible only with a simple illuminator source and optics components. Excimer laser sources, and larger, more complex illuminators, are unpractical to attach to the metroframe and are hence connected to an external frame.

In Figure 4, the table holding the wafer is called the mirror block because of the mirroring side surfaces, which allow interferometric position measurement (IFM). The interferometer is connected to a sensor plate, which is fixed to the metroframe. Assuming that both the lens and the sensor plate are rigidly attached to the metroframe, the interferometer position measurement is equivalent to a relative position measurement of the mirror block with respect to the projection lens.

Sensors that measure the height and tilt of the wafer surface under the projection lens are also connected to the sensor plate. These sensors are required to be able to keep the wafer surface in the focal plane of the lens. Note that the unflatness of the raw wafer, as well as the presence of existing layers in the chip under construction, necessitate the wafer height measurement and control.

In Figure 4, the mirror block is supported by an air foot. The air foot enables the mirror block to float on a layer of air on top of a polished granite stone. The pressurized air lifting the air foot is brought into the structure by means of hoses, which are not shown in the figure. It is possible that, in addition to pressurized air, some compartments in the air foot are under vacuum to increase the vertical stiffness with which the air foot is connected to the stone. Horizontally, the air foot allows frictionless motion.

To enable horizontal motion of the mirror block, motors are constructed in an H-configuration. The wafer stage, consisting of a mirror block and an air foot, is guided horizontally by roller bearings over the motor-stator surface. The motors then act both as actuator and guide, as is illustrated in Figure 5.

One linear motor drives the stage in the x direction, while simultaneously acting as a guiding beam for a roller bearing. The x motor itself can be moved in the y direction as well as in a rotational direction θ around the z axis, by means of two linear y motors. These motors also act as a guiding beam for the bearings of the x-motor stators. The

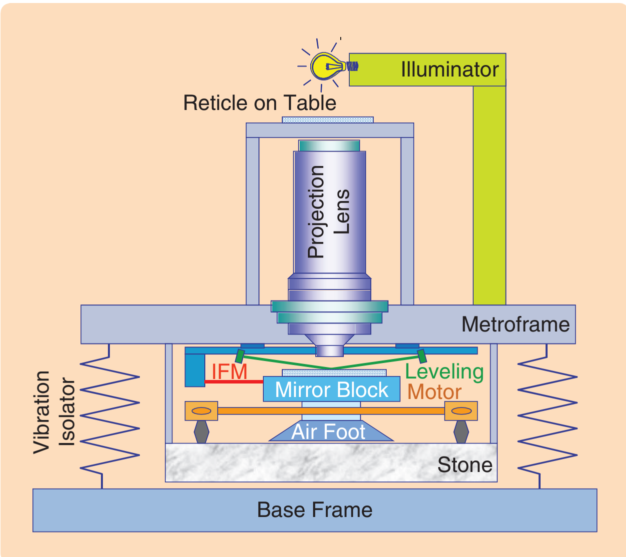

FIGURE 4 Stepper layout. The stepper exposes the complete wafer by stepping the wafer from one exposure location to the next. Light is switched off during the stepping motion. The wafer, positioned on the mirror block, is driven by an H-configuration of linear actuators, connected to the stone. The stone, lens, and sensor plate are all attached to the metrology frame, which is isolated from external vibrations by means of vibration isolators.

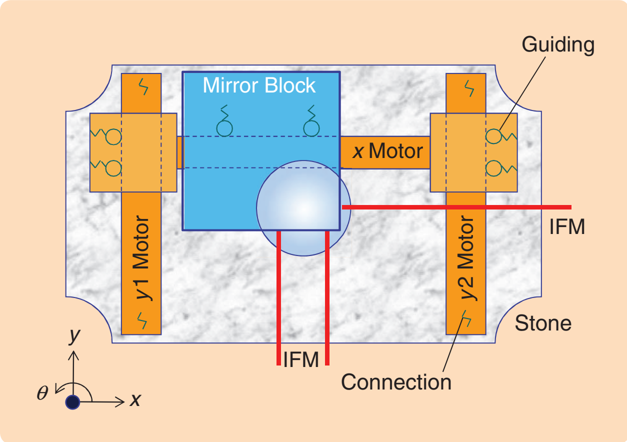

FIGURE 5 Wafer stage top view. The wafer table is driven by three actuators, two of which drive the stage in the y direction and the remaining one in the x direction. The linear actuators also function as a guide for the horizontal motion, cooperating with ball bearings in the mover. The stage floats over a granite stone by means of an air bearing, and the linear actuators are connected to this stone as well. Interferometers measure the stage position, making use of mirroring side surfaces of the stage.

two y motors are connected to a frame (not shown) that itself is connected to the stone by means of a viscoelastic material with large damping. The stone is hanging under the metrology frame, to which it is connected by leaf springs.

Vertical actuators, which are mounted between the air foot and mirror block, allow the mirror block to be moved in z direction, as well as in rotational directions around the x and y axes, called χ and ψ, respectively.

Dynamics and Control Aspects

A basic control scheme for the horizontal directions x, y, and θ is shown in Figure 6. The stage-position measurement in these directions is performed by interferometers, measuring in a lens coordinate system defined by the focal point under the lens center. In Figure 5, it can be seen that the interferometer beams are already directed toward the lens center. Because the beams measure the position of the mirror block somewhat below the wafer surface, a position correction needs to be applied based on rotations χ and ψ for which purpose additional interferometer beams are present at a lower height. This correction is called the Abbe arm correction.

To move the stage, an electric current is applied through the two y motors and the x motor. Translation of control forces and torque F_x, F_y, and T_z to individual actuator currents is needed to minimize crosstalk between motion directions.

First, to move the stage in the y direction, the correct balance between the forces of motors y_1 and y_2 depends on the actual x position. If $x < 0$, as in Figure 5, motor y_1 requires a higher current than motor y_2. Let l denote the distance between the two y-motor stators. Forces F_{y1} and F_{y2} of the two motors result in a net force F_y and a torque T_z on the stage depending on the actual x position with respect to the lens center

$$F_y = F_{y1} + F_{y2}, \tag{6}$$

$$T_z = -F_{y1}\left(\frac{l}{2} + x\right) + F_{y2}\left(\frac{l}{2} - x\right). \tag{7}$$

Inverting the corresponding matrix results in the required distribution of the two y-actuator forces

$$\begin{pmatrix} F_y \\ T_z \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -\left(\frac{l}{2} + x\right) & \left(\frac{l}{2} - x\right) \end{pmatrix} \begin{pmatrix} F_{y1} \\ F_{y2} \end{pmatrix}, \tag{8}$$

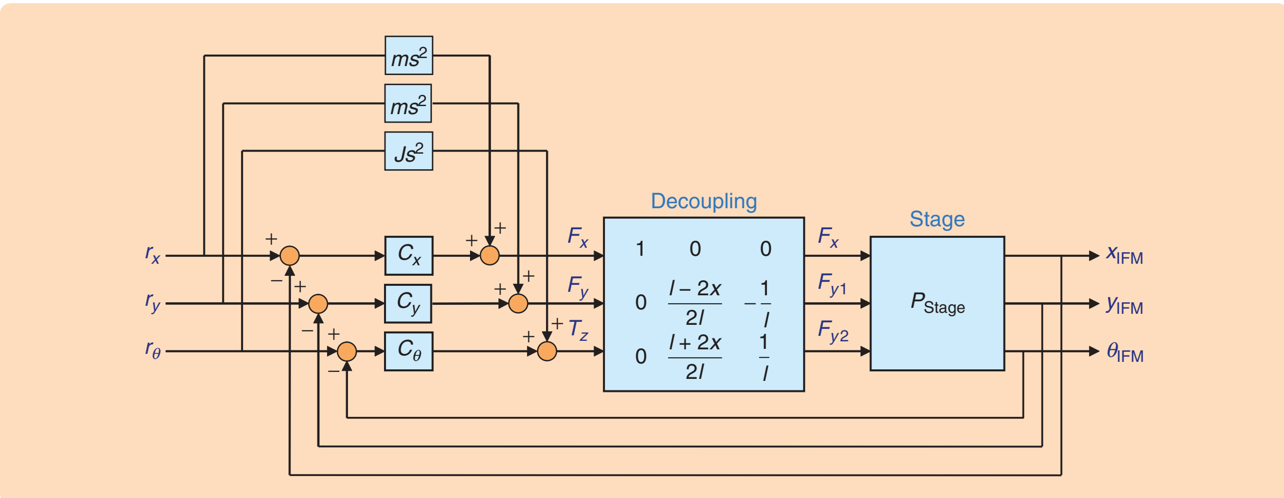

FIGURE 6 The basic horizontal controller structure. The stage positions x, y, and θ are measured by means of the interferometer system. The origin of the interferometer coordinate system is in the focal plane under the lens center since that is where the exposure takes place. Three single-input, single-output controllers, one for each direction, calculate the required force in these directions. A decoupling matrix distributes the forces over the three actuators. Note that this decoupling matrix depends on the x position. Feedforward forces are calculated by multiplying the setpoint reference acceleration by the stage's mass and inertia, respectively.

and hence

$$\begin{pmatrix} F_{y1} \\ F_{y2} \end{pmatrix} = \begin{pmatrix} \dfrac{l-2x}{2l} & -\dfrac{1}{l} \\ \dfrac{l+2x}{2l} & \dfrac{1}{l} \end{pmatrix} \begin{pmatrix} F_y \\ T_z \end{pmatrix}. \tag{9}$$

In the x direction, the force F_x is applied to the center of mass of the stage, and thus a required acceleration in x requires only the force of the x motor. In Figure 6, (9) is shown in the block indicated by decoupling.

Second, a rotation setpoint in the θ direction is defined with respect to the lens center. Applying a torque would rotate the stage with respect to its center of mass. To rotate around the lens center instead, an additional force in x or y direction is needed. This form of decoupling, linking stage and lens coordinates, is called gain scheduling and is not shown in Figure 6 but is described below in more detail.

The total transformation from forces in the coordinates (F_x, F_y, T_z) with respect to the lens, to individual actuator forces, combined with a measurement system measuring the position in these same coordinates, is required to allow the use of three independent SISO controllers. Each controller in essence encounters a mechanics transfer function of $1/(ms^2)$. However, this ideal transfer function is compromised by three phenomena.

First, an actuator force in the y direction moves the x motor stator, which in turn accelerates the mirror block. This force is exerted through the roller bearings, which are indicated as springs in Figure 5. Dynamically, the exertion of the force through these springs creates a resonance.

Second, an actuator force also results in a reaction force on the motor stator. Because the motor stators are connected to the stone forming part of the isolated metrology frame world, the reaction force excites resonances in the frame and possibly the lens. A basic dynamic model is shown in Figure 7. Here, finite stiffnesses are indicated by springs, and the corresponding resonance frequency is indicated.

The resonances in the frame and lens that are excited by the stage motion result in vibrations, which are observed in the stage control loop. Figure 8 shows the stage action path transfer function and the stage reaction path transfer function. Although in general the stage reaction path transfer function has a much smaller amplitude than the stage action path transfer function, at some resonance frequencies the reaction path impacts the combined transfer function such that the reaction path limits the obtainable stage control bandwidth.

Third, the roller bearings provide almost frictionless movement over longer distances but tend to stick to the guidings when movement is small. In the nanometer (and micrometer) region, the bearings act as springs with unknown stiffness, resulting in a considerably varying transfer function, as shown in Figure 9. Mechanical tolerances of the guiding beams and rollers, pretension, and

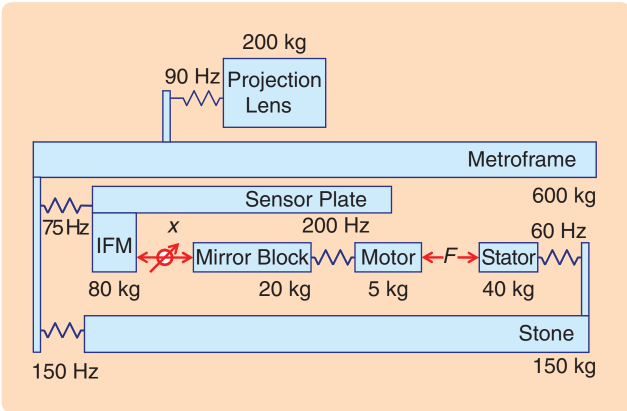

FIGURE 7 Horizontal 1DOF dynamics. This diagram shows a simplified dynamic model of the machine, with each stiffness being shown as a spring. A motor force F is exerted on the motor body, which is connected to the mirror block with a stiffness corresponding to a 200-Hz resonance. The reaction force is exerted on the motor stator, which is connected to the stone with a 60-Hz resonance. The stone, metroframe, sensor plate, and lens are all connected with limited stiffness to each other. Although this model is 1DOF only, its dynamic properties can be used to determine attainable performance and possible structural improvements.

lubrication form relevant factors in limiting this friction-induced stiffness.

As an example, a 30-Hz bandwidth proportional integral differential (PID) controller with lowpass filter is applied, with the transfer function

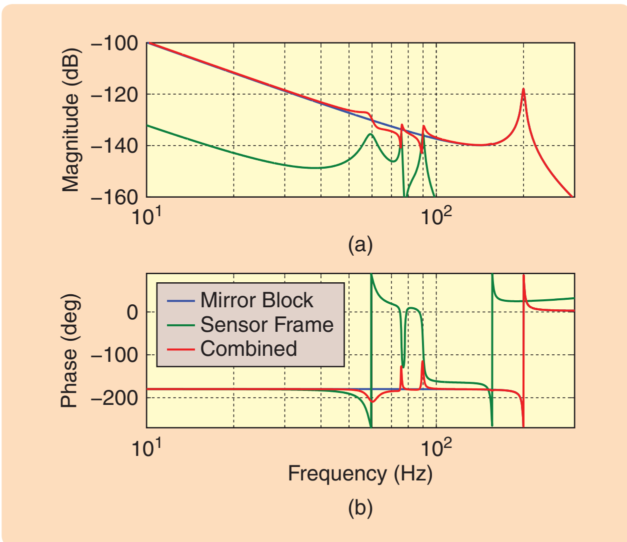

FIGURE 8 Mechanics transfer function from the force F to the measured position x. The action force on the stage results in a motion described by a mass line with one resonance at 200 Hz, created by the motor-to-mirror block coupling (blue). The reaction force excites the flexible modes of the components in the system, leading to motion of the interferometers mounted on the sensor plate (green). When the magnitude of this reaction path is similar to that of the action path, the reaction path impacts the transfer function as observed by the controller (red) and limits the attainable bandwidth.

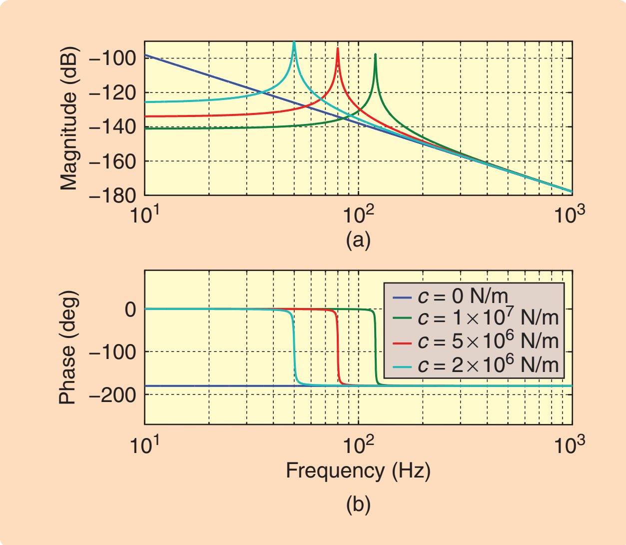

FIGURE 9 Transfer function from F to x with and without bearing stiffness. Without bearing stiffness, a mass line with a -40 dB/decade slope is present, corresponding to the 20-kg mass. Note that the motor and its coupling are disregarded in this plot. For small movements, the bearings behave as a stiffness because the roller balls stick to the guiding beam, depending on friction. The stiffness, and hence the resonance frequency, is not constant. As a result, the transfer function as observed by the controller can be any one of the shown transfer functions at a specific time. By keeping the maximum occurring stiffness low, for example by lubrication or selecting a proper bearing pretension, the resonance frequency always stays below a maximum value. If this maximum resonance frequency is smaller than the stage bandwidth, stability can be obtained.

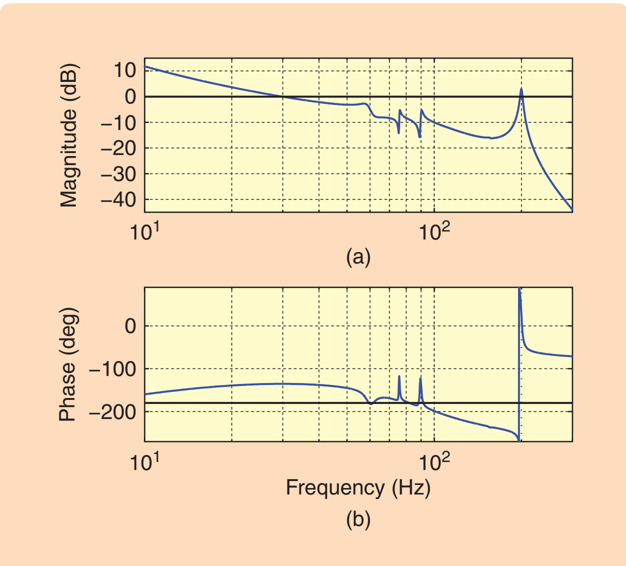

FIGURE 10 Bode plot of the open-loop transfer function with a 30-Hz bandwidth controller. The 200-Hz resonance is actively damped by creating a favorable phase by means of a lowpass filter. At 60, 75, and 90 Hz, frame resonances are visible, with the 60-Hz resonance being caused by the compliance between the actuators and the stone. Although the 60-Hz resonance is damped, this resonance degrades the stability properties.

TABLE 2 Proportional-integral-differential parameters. Proportional gain, integrator and differentiator frequencies, and lowpass frequency are all coupled with the required bandwidth by the factor α. A small value of α results in a small phase margin but a high low-frequency gain. A large value of α does the reverse. A practical value for α is 3.

Name	Meaning	Nominal value
f_{bw}	Desired bandwidth (Hz)	
m	Mass to be moved (kg)	
α	PID frequency ratio	
K_p	Proportional gain	$m(2\pi f_{bw})^2/\alpha$
f_i	Integrator frequency	f_{bw}/α^2
f_d	Differentiator frequency	f_{bw}/α
f_{lp}	Lowpass filter frequency	αf_{bw}
z_{lp}	Lowpass filter damping ratio	$z_{lp} \in [0.5,\ 1]$

$$H_{\mathrm{ctrl}}(s) = K_p\left(\frac{s+2\pi f_i}{s}\right)\left(\frac{s}{2\pi f_d}+1\right) \times \left(\frac{(2\pi f_{\mathrm{lp}})^2}{s^2+2z_{\mathrm{lp}}(2\pi f_{\mathrm{lp}})s+(2\pi f_{\mathrm{lp}})^2}\right). \tag{10}$$

The PID controller tuning is of a standard type in which the ratio between the desired bandwidth f_{bw}, the integrator frequency f_i, the differentiator frequency f_d, and lowpass filter parameters f_{lp} and z_{lp} is determined by the factor α, as listed in Table 2. In the example here, $\alpha = 3$ and $z_{\mathrm{lp}} = 0.7$. A larger value of α creates a better phase margin but reduced disturbance rejection at low frequencies. Figures 10 and 11 show the open-loop Bode and Nyquist plots, respectively. Figure 11 shows that, due to the frame resonances, especially around 60 Hz, a larger loop gain could create instability.

In addition to this feedback controller, a feedforward controller is present, as shown in Figure 6. The setpoint acceleration, obtained by double differentiating the reference position, is multiplied by the stage mass m and inertia J, and added to the controller output. This feedforward force takes care of stage motion without the need for a controller input, which equals the position error. Note that digital differentiation of a smooth position reference is not a problem and that setpoint design itself is an interesting topic that deserves attention.

Figure 12 shows the time response of the stage to a position reference change, with a limited setpoint acceleration of 10 m/s^2. Figure 12(a) shows a 20-mm stage motion setpoint and actual position, together with a scaled acceleration. On this scale, the stage follows its setpoint accurately. Figure 12(b) shows the controller error, which is in the 50-μm range during the move and decreases only slowly after that. The imaging error, defined as the position deviation of the image on the wafer, corresponds to

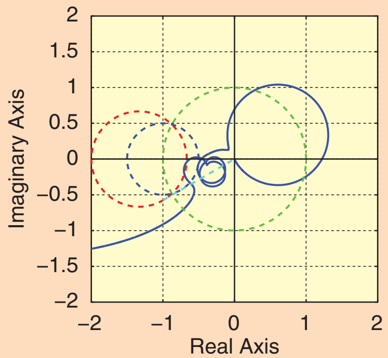

FIGURE 11 Nyquist plot of open-loop transfer function with a 30-Hz bandwidth controller. This plot shows the active damping of the 200-Hz mode, which is in the right-half complex plane. The 60-Hz resonance makes a direct turn toward the point –1, endangering stability and limiting the possibility of obtaining a high control bandwidth.

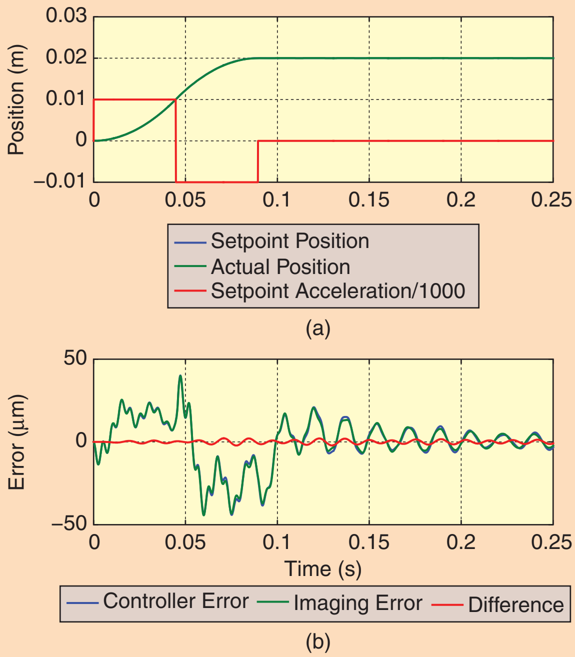

FIGURE 12 Step move of 20 mm. (a) shows the acceleration reference (red) and the setpoint position (blue). The actual response, shown in green, matches the position reference. (b) shows the controller error (blue) and the position error of the image on the wafer (green), which resemble each other. This resemblance means that the controller error is a useful indication of the actual image error, and the interferometer and lens are sufficiently coupled dynamically.

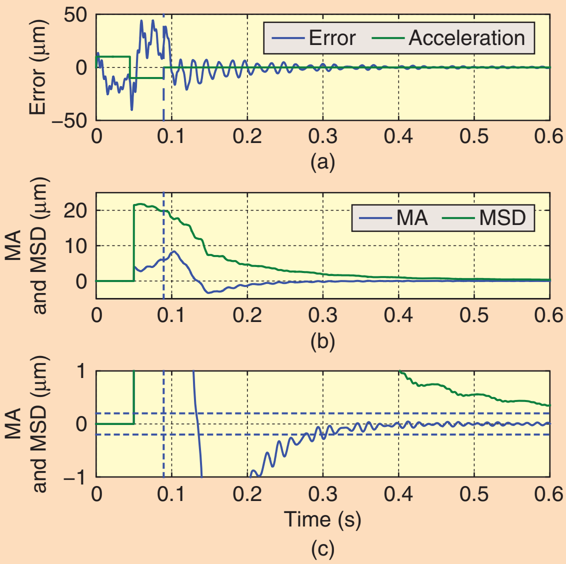

FIGURE 13 The moving average (MA) and moving standard deviation (MSD) of the controller error. This plot shows the MA and MSD, relevant for overlay and imaging requirements, progressing over time. Part (a) shows the acceleration profile (green) and controller error (blue). The dashed line indicates the end of the step setpoint. Part (b) shows the MA and MSD values (blue and green, respectively), and (c) shows the same values at a smaller scale. The horizontal dashed lines indicate the 200-nm error limit allowed for the particular exposure process.

the controller error. The remaining difference is caused by the remaining lens vibration and sensor-plate vibration, excited by the reaction forces of the stage. At these frequencies the stage's disturbance rejection is mostly absent because of its 30-Hz bandwidth.

A similar situation holds in the vertical direction. Vertical forces induced by leveling are exerted on the stone through the air foot. Dynamically speaking, the air foot can be considered a spring whose stiffness and damping depend on air pressure, vacuum level, and bearing design. The vertical forces end up in the system as much as the horizontal forces, creating similar frame-resonance effects.

Stepper Imaging

Even in the presence of the above-mentioned limitations, the system can function thanks to a key advantage. After stepping the wafer to a new position, the wafer stage is allowed to wait until its position has settled such that the remaining error is sufficiently low before switching on the illuminating light. The MA and MSD after the step motion indicate the usability of the system for imaging.

Figure 13 shows these values as a function of time, using a 100-ms exposure time window. Figure 13(a) shows the raw servo error and scaled acceleration setpoint that ends at $t = 0.09$ s. Figure 13(b) shows the MA and MSD values as functions of time. Note that the first relevant MA and MSD value can be calculated after 100 ms. This value is located at $t = 50$ ms in Figure 13 due to the MA and MSD definitions in (4) and (5), respectively. The first MA and MSD values relevant for imaging can be read at $t = 0.14$ s, 50 ms after the end of the acceleration phase. Figure 13(c) shows the

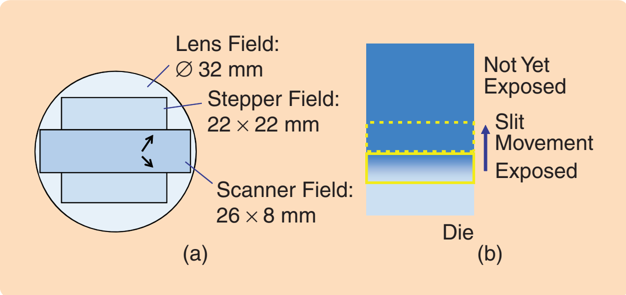

FIGURE 14 Stepper and scanner lens field. While the lens usually has a circular field, a chip is rectangular in shape. A 32-mm lens field allows imaging of a 22 × 22-mm chip. Making larger chips requires a larger lens, which increases tool cost. Instead, a narrow slit-like field is used while making use of the same lens. The large field size is then created by scanning the wafer through this illuminated area. An extra advantage is the averaging effect of lens aberrations (indicated by arrows), because each exposed point in the die is the result of scanning through the lens field.

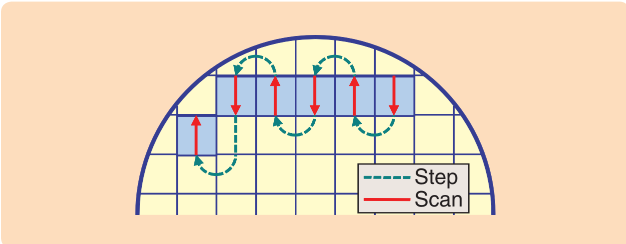

FIGURE 15 The step-and-scan principle in exposing a wafer. The wafer makes a scanning motion under the projection lens (red arrows). After the scan finishes, the light source is switched off, and a stepping movement is made to the start of the next image field (green, dashed arrows). Since the reticle has to scan synchronously with the wafer, it is advantageous to meander the wafer to avoid throughput loss.

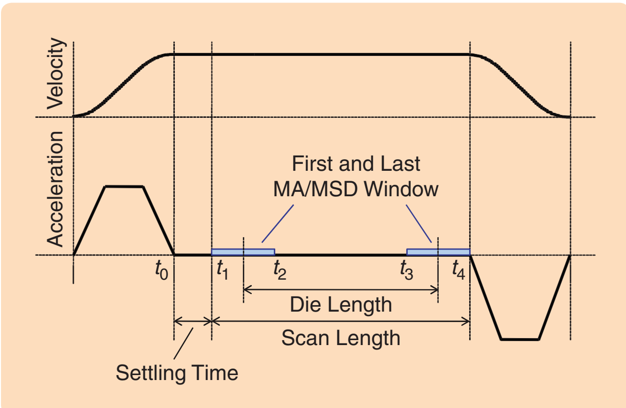

FIGURE 16 Scan profile and moving average (MA)/moving standard deviation (MSD) windows. After accelerating the wafer and the reticle, a settling time is allowed to reduce positioning errors. The position error that each point in the die experiences during its exposure is expressed in MA and MSD values. The first point in the die is exposed in the time interval $[t_1, t_2]$, and its MA and MSD values are calculated from the position error in this interval. The total constant-velocity scan length then equals the die length, plus the slit height, plus the distance moved during the settling time.

values on a finer vertical scale. Dashed specification lines at ±200 nm are drawn. It can be seen that if a maximum MA of 200 nm is allowed by the imaging process, exposure can start at $t = 0.3$ s. The corresponding MA and MSD values, calculated over the interval $t = [0.3, 0.4]$ s, are shown at $t = 0.35$ s. The MSD value at that time equals 2 μm; if this value is incompatible with the imaging process a longer settling time is required.

THE SCANNER

The normally rectangular form of a chip is exposed on the wafer using a normally circular projection lens. As such, a 32-mm-diameter lens field can expose square dies of 22 × 22 mm. To make larger ICs, a larger lens is required, which, however, is prohibitively expensive. The solution to this problem is to expose a chip not in one flash but in a scanning fashion. Now, a rectangular lens field is used that can be wider than before, usually 26 mm. In the scanning direction, the lens field is much smaller, such as, for example, 8 or 10 mm, as visualized in Figure 14.

Instead of the step-and-expose principle described above, in the step-and-scan method, each exposure is created by scanning the wafer through the illuminated narrow slit. The new expose field is illustrated in Figure 14. The length over which the wafer scans defines the height of the die, which usually equals 32 mm. Hence, by using the step-and-scan method, a larger image can be created with the same projection lens. This lithographic machine, which is called scanner, was brought to market around 1997 [6]. The major implications in the transition from stepper to scanner are described below.

To expose the full image, not only the wafer but also the reticle needs to make a scanning movement. Because the magnification factor of the projection lens is usually 1/4, the reticle needs to scan with a scan speed that is four times higher than that of the wafer. The motion of the reticle, the motion of the wafer, and the dose control of the illumination laser need to be tightly synchronized. Other components in the tool may need synchronization as well, such as reticle masking and lens manipulators.

A high positioning accuracy is now required during the scanning motion instead of during standstill. This requirement, for example, prohibits the usage of roller bearings as used in the stepper. In addition, it is no longer possible to wait until the system is settled and then start the exposure. When wafer and reticle have accelerated to the start location of a die, both must be at the correct position with the required accuracy because waiting longer would result in an incompletely exposed die.

Scanning Performance Measures

Figure 15 further illustrates the step-and-scan principle in exposing part of a complete wafer. The straight red arrows indicate synchronous constant-velocity motion of the wafer and the reticle, gradually exposing each die as described above. The dashed lines, representing wafer movements, are steps that bring the stage to a new start location for the next die. During these steps, the light source is switched off.

Figure 16 shows a more detailed timing diagram of the stage movement during a scan. Figure 16(b) shows an acceleration setpoint profile, which in this example is a third-order profile instead of the previously used second-order profile. Figure 16(a) shows the velocity setpoint profile. At $t = t_0$, the acceleration phase ends, and a constant velocity is reached. After a certain settling time, which allows the remaining controller error to be reduced to an acceptable value, the first point in the die to be exposed enters the illumination slit at $t = t_1$. At $t = t_2$, this first point on the die leaves the slit again. The stage-positioning errors in the interval $[t_1, t_2]$ determine the effect on overlay and imaging. Hence, the calculated MA and MSD values over this first interval correspond to the effect of positioning errors on the first point in the die. At $t = t_3$, the last point of the die enters the slit, and, finally, at $t = t_4$ the die leaves the slit again, making the interval $[t_3, t_4]$ the last window over which MA and MSD values need to be calculated. Hence, the total scan length of the stage equals the length of the die, plus the height of the slit, plus the length needed for settling of the stage. After $t = t_4$, the stage decelerates again to standstill or follows another trajectory that brings the stage to the start of the next die.

Scanner Dynamic Architecture Changes

To allow nanometer-range positioning errors during the scan, the reaction forces need to be led away from the metrology frame or its connected components. Figure 17 shows the modified system. The stage actuators are no longer connected to the stone but are now mounted to an external base frame. The metrology frame is again suspended from the base frame by isolators. The reticle is also positioned on a movable stage.

By connecting the linear actuators to the base frame, the risk arises that base frame vibrations, for example induced by the floor to which the base frame is attached, disturb the stage. By selecting Lorentz actuators for driving the stage, a natural decoupling of base-frame vibrations to the stage is guaranteed. Since the force acting on the stage depends linearly on the actuator current only and is independent from the stator motion, external vibrations do not result in forces on the stage. A truly linear Lorentz actuator, however, has a range of only about 1 mm, which is insufficient for the stage that needs translation of at least 300 mm. This problem is solved by introducing a second actuator, which can travel over large distances with moderate micrometer-range accuracy, and moves the stator part of the short-stroke Lorentz actuator. Usually the coil part is connected to the long-stroke motor to avoid crossover of cables and cooling hoses to the accurate short-stroke system.

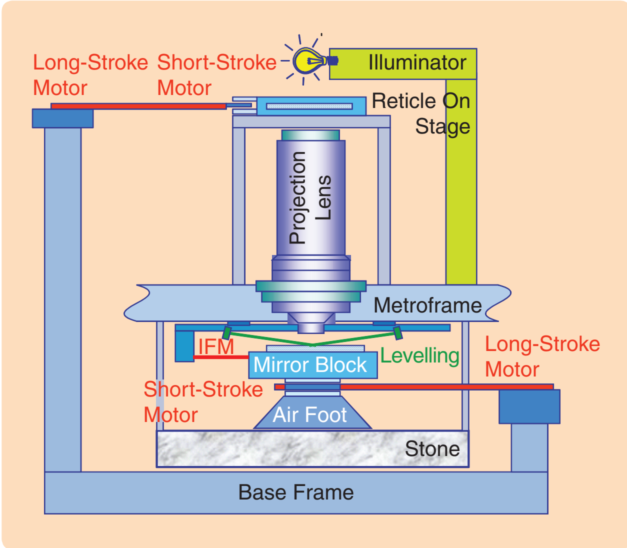

FIGURE 17 Scanner layout. The scanner differs from the stepper mainly in the presence of a reticle stage that can move the reticle in synchronization with the wafer, and the connection of the stage actuators to the base frame instead of the stone, leading the reaction forces away from critical components such as the lens and measurement systems. Small positioning errors with respect to the setpoint now need to be achieved during the constant-velocity scan, requiring changes in actuator concepts, guiding, and motion controller timing. Leveling, that is, keeping the wafer in the focal plane of the lens, now has to be performed on the fly, requiring coupling between the horizontal and vertical controllers.

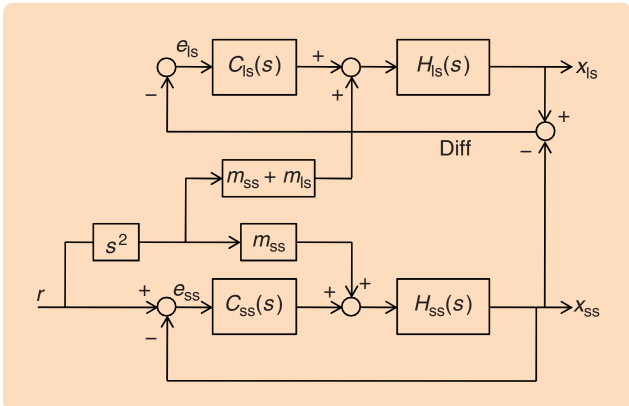

FIGURE 18 Long-stroke control. The long stroke carries the stator part of the short-stroke motor, typically the coils. Its main goal is to keep the coils as close as possible to the magnets to avoid linearity errors in the short-stroke Lorentz actuator. Therefore, the position difference between the long stroke and short stroke serves as the controller input. The long-stroke behavior as observed by the controller is independent of that of the short stroke because the latter is controlled with respect to the metroframe. However, the feedforward force needs to account for the reaction force of the short stroke.

Figure 18 shows a basic control scheme that combines the short-stroke and long-stroke stage controllers. The control loop containing $H_{ss}(s)$ represents a 1DOF short-stroke axis, similar to one of the control loops in Figure 6. The control

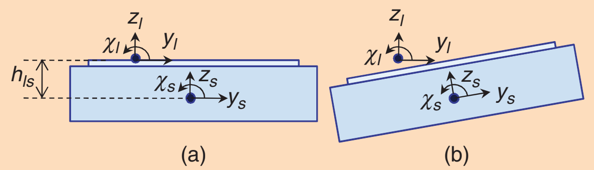

FIGURE 19 Vertical lens and stage coordinate systems. A torque T_x on the stage results in a rotation χ about the x axis. Such a rotation leads to a z error of the wafer at the point of exposure, depending on the y position. In addition, a y error is created because the center of mass is a distance h_{ls} below the wafer surface. Additional forces in the z and y directions compensate this effect.

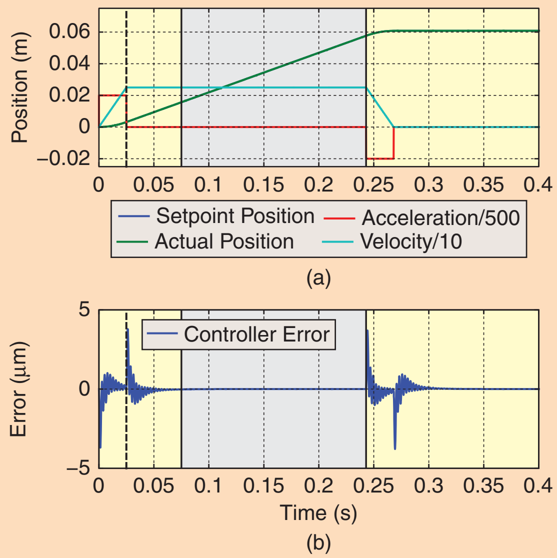

FIGURE 20 Scanning motion with the ideal system. (a) shows the setpoint position (blue), scaled acceleration (red), velocity (cyan), and the actual position (green). (b) shows the controller error during the exposure time window, dominated by a 400-Hz settling effect due to the motor-to-mirror block connection.

TABLE 3 Scan parameters. These typical values relate to the scanner, which exposes the die in a scanning fashion. In the step movements from die to die, higher values for the speed may be used. Dual-stage scanners use higher accelerations of up to 40 m/s^2 and scan speeds up to 0.6–1 m/s for the wafer stage.

Parameter	Value
Slit height	10 mm
Scan speed	0.25 m/s
Acceleration	10 m/s^2
Settling time	50 ms
Die length	32 mm
Total scan length	32 + 10 + (0.25)(50) = 54.5 mm

loop containing $H_{ls}(s)$ represents the long-stroke control system. Each axis can be considered separately because of decoupling of the control forces.

The long-stroke system is required to track the short-stroke system to keep the coils and magnets of the short-stroke Lorentz actuators close together. To this end, a sensor measures the relative position of the long stroke with respect to the short stroke, indicated by the "diff" signal in Figure 18. This difference is used as input for the long-stroke system controller. The position x_{ss} of the short stroke can be regarded as a position reference for the long-stroke system control loop. The long-stroke controller $C_{ls}(s)$ is designed for the mass and dynamics properties of the long-stroke system without taking the short stroke properties into account. In the long-stroke, open-loop transfer function, the short-stroke system is not visible because the short stroke tracks the interferometer and is decoupled from the long stroke.

In the feedforward path, however, the mass of the short-stroke system must be included since a short-stroke acceleration adds reaction forces to the long stroke in addition to the forces needed to accelerate the long stroke.

For wafer leveling, the actuators drive the mirror block with respect to the air foot, and hence vertical reaction forces can directly enter the silent, vibration-free, metro-frame world. Leveling now needs to be performed during scanning, making use of the wafer-height measurement by the level sensor. Some coupling between the horizontal and vertical control loops is now required. Figure 19 shows the result of a torque T_x around the x axis, namely a stage rotation around its center of mass. The rotation χ results in an undesired shift in the y direction by the amount $h_{ls}\chi$. By adding the force $F_y = h_{ls}(m/J_x)T_x$ to the y-controller output, this shift can be corrected. Another effect is defocus, and thus an extra force F_z is needed. These steps represent the progression from completely independent horizontal and vertical controllers in the stepper to an integrated 6DOF control system.

By leading away the reaction forces, the reaction path as shown in Figure 8 is reduced and is no longer a factor in the stage's controller design. This advantage allows for larger bandwidths and correspondingly smaller controller errors. The motor-to-mirror-block coupling plays the same role as above but is more high frequency because its stiffness can be much higher than that of roller bearings. This higher stiffness increases the coupling frequency from 200 to 400 Hz, and the bandwidth increases correspondingly from 30 to 80 Hz. Figure 20 shows a typical example of time-domain behavior. Here, the scan parameters in Table 3 are used. Figure 20(a) shows the acceleration, velocity, and position profile as functions of time; Figure 20(b) shows the raw controller error during the move. Vertical dashed lines indicate the end of the acceleration phase ($t = 0.025$ s). The scanning period between the end of the settling phase ($t = 0.075$ s) and the end of the constant-velocity phase

($t = 0.243$ s) is indicated by a light gray area. After settling, the total scan length equals 42 mm, consisting of 32-mm die length and 10-mm slit height. Some vibrating behavior is visible in response to setpoint acceleration changes. This behavior can be attributed to a discrepancy between the used acceleration feedforward and the actual inverse of the process transfer function, which includes the motor resonance of 400 Hz.

Figure 21 shows the MA and MSD result of the scan. Because the slit height in this example equals 10 mm and the scan speed equals 250 mm/s, the time window for the MA/MSD calculation equals 40 ms. This window is the time span that each point on the die spends in the illuminated slit. Hence, the first useful MA/MSD value occurs at a time of 20 ms after the settling phase. Figure 21(a) shows the raw controller error and a scaled version of the setpoint acceleration; the vertical dashed line coincides with the end of the acceleration phase. The scan part is again indicated by a light gray area. Figure 21(b) shows the MA/MSD values, with the gray area indicating the relevant values being bound by the first and last points in the die. Figure 21(c) shows a magnification of (b). It can be seen that at the start of the die, MA and MSD are both approximately 10 nm. Hence, with the system described above, imaging requiring MA and MSD values of 10 nm can be performed.

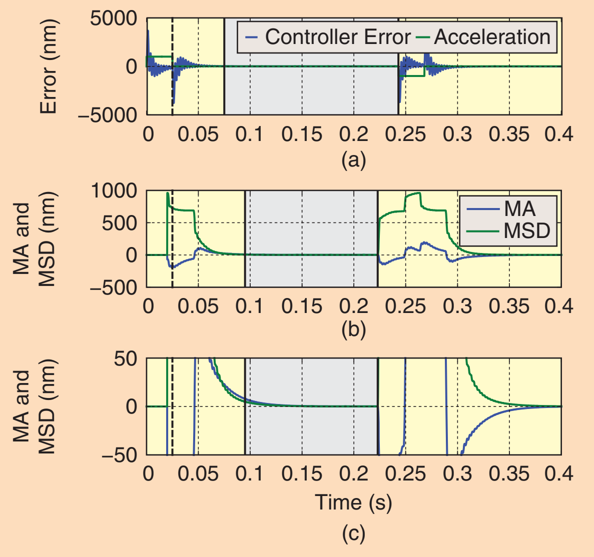

FIGURE 21 Moving average (MA) and moving standard deviation (MSD) during scan for the ideal system. The controller error of Figure 20 is copied in (a), while (b) and (c) show the MA and MSD values. These values are valid over the die length, while the raw error is relevant over the total scan length, excluding settling time. This valid range is related to the fact that the scan length equals the die length plus the slit height.

THE DUAL-STAGE SCANNER

Around the year 2000, the semiconductor manufacturing industry required a wafer size change from a 200-mm to a 300-mm diameter, enabling placement of more than twice the amount of chips on each wafer. Simultaneously, the chip design shrink continued, demanding higher positioning accuracies for the stages. This change meant that the stages had to become larger to support 300-mm wafers, had to become faster to keep the number of wafers per hour high enough, and had to become more accurate to enable smaller minimum feature sizes.

With the scanner layout described in the previous section, the combination of these elements proved impossible. A new architecture had to be created that induced even smaller disturbances on the metrology frame by the high-speed moving stages and vertical leveling forces. Furthermore, to allow the use of feedforward to reduce leveling errors, the wafer surface height map needed to be measured before exposure. Finally, a reduced overhead time created by wafer exchange and wafer alignment was needed to keep the machine throughput high.

A solution was found by equipping the system with two wafer stages [7]. While the first stage exposes the wafer, the second stage unloads the previous wafer from the tool, loads a new wafer on the stage, aligns the horizontal placement of the wafer on the stage, and measures the wafer height map used to focus the wafer during exposure. When both stages are finished with their tasks,

FIGURE 22 Current model dual-stage scanner. ASML's TWINSCAN NXT:1950i dual-stage scanner processes 175 wafers per hour, using a 193-nm ArF excimer laser as its source. Using immersion technology allows exposure of patterns of 38 nm, close to the theoretical limit of 36 nm. Targeted for double-patterning exposures, its overlay specification is 2.5 nm. High throughput is achieved by dual planar actuators as long-stroke motors, reducing stage swap time. Replacing interferometers by large, planar encoders allows reduction of overlay values to below 4 nm, by reducing sensitivity to environmental conditions such as air pressure and temperature.

the stages are swapped and a new cycle begins. In this way, the number of wafers that is processed is enlarged by removing overhead time from the expose cycle. The increased stage acceleration and speed further improves

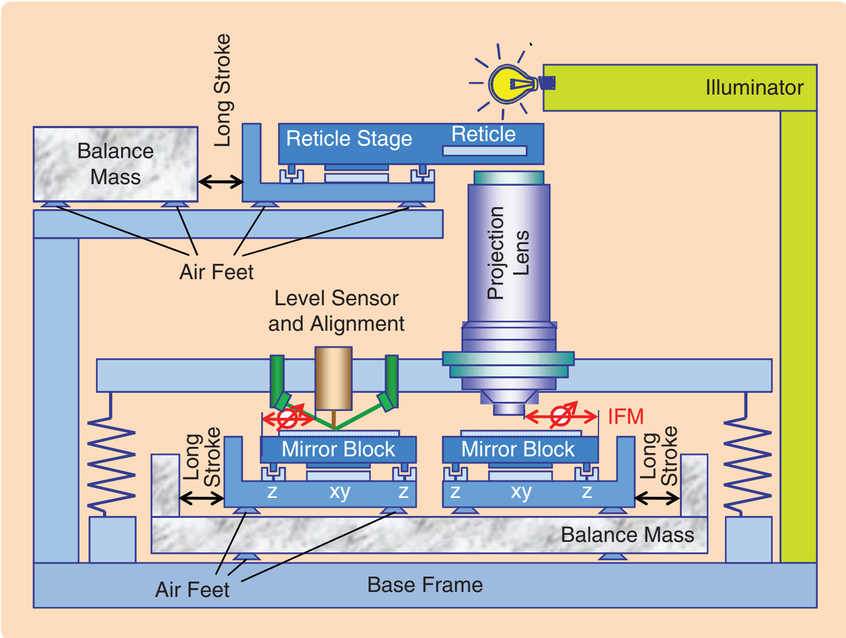

FIGURE 23 Dual-stage scanner layout. The dual-stage scanner uses two wafer stages, enabling simultaneous measurement of the wafer-surface-height map and horizontal alignment, as well as exposure of the previously measured wafer. This sequence reduces overhead time while facilitating extended wafer alignment. Reaction forces are now led to balance masses to further reduce disturbance forces on the metrology frame.

throughput. Figure 22 shows a commercially available lithographic tool, while Figure 23 shows a dynamic layout of a dual-stage scanner.

Dual-Stage Scanner Architecture Changes

A major change in the dynamic architecture is the use of balance masses [8]. To avoid any remaining part to be transported toward the metroframe due to nonperfect isolation, horizontal reaction forces are no longer exerted on the base frame. Now, the long-stroke motor stators are connected to the balance mass, which is allowed to move freely in horizontal directions by means of air bearings. A force on the stage results in a motion of the balance mass in the opposite direction, proportional to the mass ratio of the stage and the balance mass. Since the mass of the balance mass is much higher than that of the stage, its motion is smaller.

To keep the balance mass from drifting, actuators are used to control the balance mass position. Figure 24 shows the control scheme. The force F_s needed to move the stage acts as a reaction force on the balance mass m_b. The positions x_s of the stage and x_b of the balance mass are used to calculate the center of gravity, x_{cog}, of the combined system. In Figure 24, x_{cog} is multiplied by $m_b + m_s$ to use the standard controller design. As long as x_{cog} is zero, the balance mass controller takes no action; only deviations lead to a reactive control force on the balance mass. Again, each coordinate can be treated separately because of force decoupling of all systems.

As shown in Figure 23, the base over which the stage floats is no longer part of the isolated metrology frame world but is now connected to the base frame. To avoid vibrations entering the mirror block, a Lorentz actuator is now also used for vertical directions, providing isolation in these directions as well. Because the required vertical range is smaller than 1 mm, no separate long-stroke motor is required. A 6DOF Lorentz-actuated block is the result.

Stage position measurement is now performed in all degrees of freedom by interferometers, with reference beams directed at the projection lens. This method provides a direct relative measurement of the position with respect to the lens. At the second stage, the wafer is loaded, and its surface is mapped in horizontal and vertical planes with respect to the stage itself. After the stage swap, the stage that is now positioned under the projection lens is aligned to the reticle in 6DOF by means of a through-the-lens optical system. With the wafer surface position known with respect to the stage and the stage position known with respect to the reticle, exposure can start.

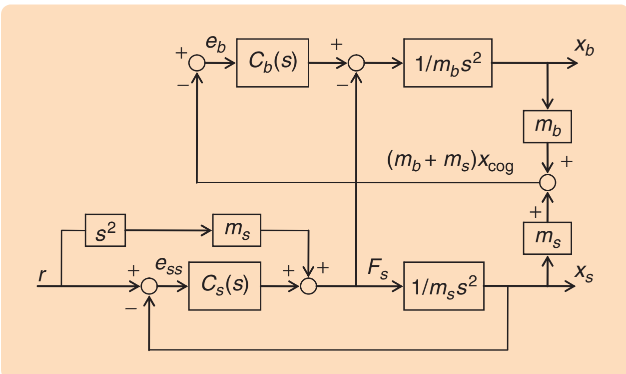

FIGURE 24 Balance mass control. The force F_s acting on the stage in the control loop involving $C_s(s)$ also acts on the balance mass as reaction force, in the control loop involving $C_b(s)$. The combined center of mass of the stage and balance mass serves as a balance mass controller input. In this way, the nominal motion of the balance mass does not result in controller action. Only deviation from this natural behavior leads to a corrective controller action. In some directions, notably θ, the balance mass behavior needs to be controlled more strictly because stage motion in the horizontal plane may result in unbounded balance mass rotation.

INTEGRATED 6DOF CONTROL OF A STAGE

We now focus on controlling a 6DOF stage equipped with full Lorentz actuation, which means that forces in 6DOF

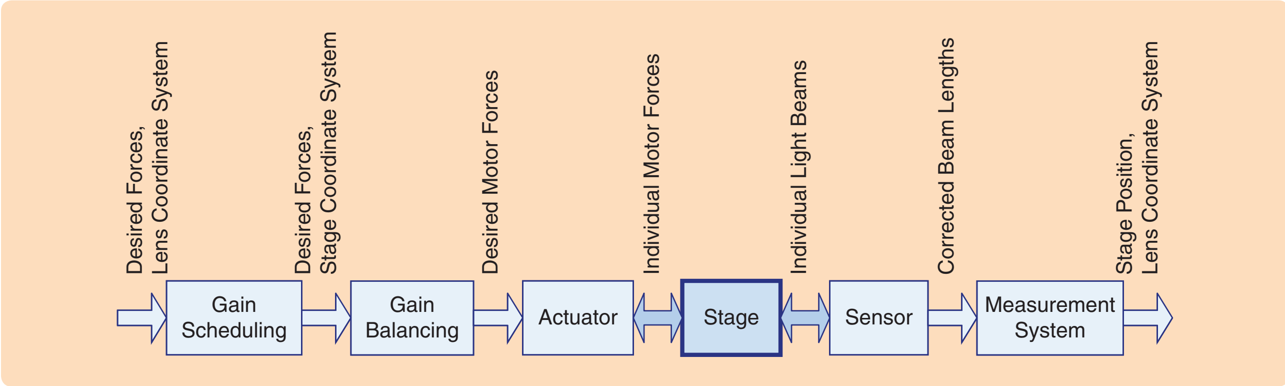

FIGURE 25 Physical plant components. The stage position is measured by a sensor, mostly by several interferometers. The beam lengths are corrected for environmental conditions and mounting tolerances by the sensor software block. From the corrected beam lengths, the stage position in lens coordinates is deducted, yielding the position of the stage in the focal plane under the lens center, where exposure takes place. The stage is driven by actuators, which also need correction for variations in the motor constant and mounting tolerances. Control forces in lens coordinates need to be converted to forces in stage coordinates (called gain scheduling), which in turn are converted into individual motor forces, a process called gain balancing.

can be applied to the mirror block. The same strategy as above is used, namely, decoupling the process behavior such that cross terms are minimized, enabling the use of 6 SISO controllers. This method relies on a high degree of stage linearity and decoupling; this section focuses on decoupling. Figure 25 shows the main components of the process as encountered by the six SISO controllers.

Process Building Blocks

The stage position is measured using several sensors, most commonly interferometers. Each measurement beam is corrected for environmental conditions such as pressure and temperature. Also, mechanical misalignment, that is, pointing error, and mirror unflatness are corrected on a beam-by-beam basis. The corrected data from all sensors is converted into Cartesian coordinates in the measurement system, providing stage positions $x_l^T = (x_l\ y_l\ \theta_l\ z_l\ \chi_l\ \psi_l)$ in the lens coordinate system, which has its origin in the focal plane under the lens center. Here, θ denotes rotation around the z axis, χ denotes rotation around the x axis, and ψ denotes rotation around the y axis.

The control forces, as determined by the six SISO controllers, are also in lens coordinates. However, the stage forces are applied by the short-stroke actuators, which have a fixed position and actuation direction with respect to the stage center of mass. The fact that the stage center of mass does not coincide with the lens coordinate system's origin implies that a conversion must take place from forces $f_l^T = (F_{xl}\, F_{yl}\, T_{zl}\, F_{zl}\, T_{xl}\, T_{yl})$ in the lens coordinate system into forces $f_s^T = (F_{xs}\, F_{ys}\, T_{zs}\, F_{zs}\, T_{xs}\, T_{ys})$ in the stage coordinate system. This step is called gain scheduling since the transformation turns out to be position dependent. The next step is then the conversion of forces in the stage coordinate system into forces $f_m^T = (F_{x1}\, F_{y1}\, F_{y2}\, F_{z1}\, F_{z2}\, F_{z3})$ for the individual motors. This step, which is called gain balancing, is completely determined by the geometric actuator layout in the stage. Here, one x actuator and two y actuators are used, as in Figure 26.

Horizontal Gain Scheduling

Figure 27 shows the stage with its position x_s and its position in lens coordinates x_l. The aim is to transform horizontal

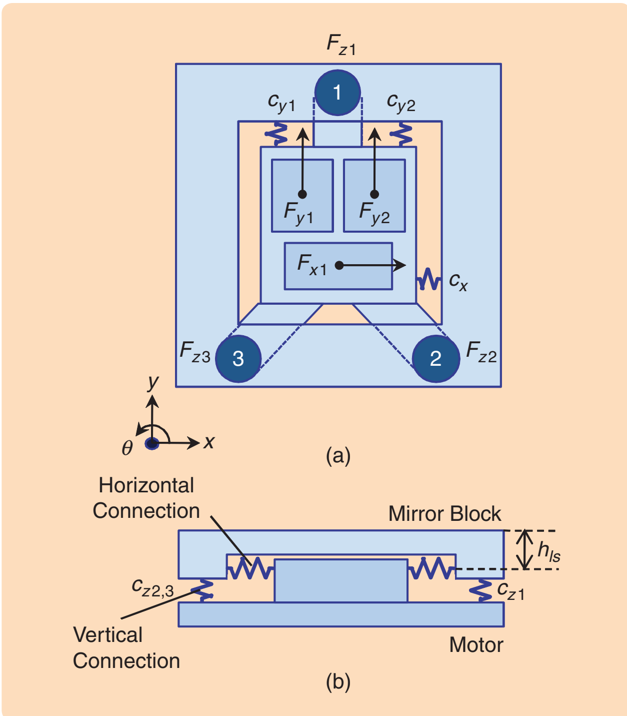

FIGURE 26 Stage layout with motor connection stiffness. The motor block, which is driven by the long-stroke motor, produces 6DOF forces. The motor block is not rigidly coupled to the mirror block, as indicated by springs, and therefore shows up as dynamic behavior in the stage transfer functions. Nonrigid coupling can be dictated, for example, by thermal requirements that require a certain amount of mechanical decoupling between the structures.

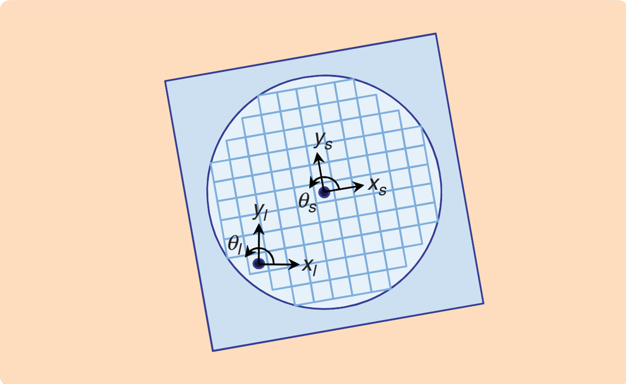

FIGURE 27 Lens coordinate system with the stage coordinate system for the horizontal directions. A rotation θ results in an undesired motion in the x or y direction of the point of interest under the lens. Position-dependent corrective forces, depending on T_z, compensate for this effect to a great extent. Additional corrective terms may further improve decoupling.

forces in the lens coordinate system to forces in the stage coordinate system using the relations

$$x_l = x_s\cos(\theta) + y_s\sin(\theta), \tag{11}$$

$$y_l = -x_s\sin(\theta) + y_s\cos(\theta). \tag{12}$$

Note that θ does not have a subscript because θ is independent of the coordinate system. In matrix form,

$$\begin{pmatrix} x_l \\ y_l \end{pmatrix} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix}\begin{pmatrix} x_s \\ y_s \end{pmatrix}, \tag{13}$$

but since x_s as a function of x_l is needed,

$$\begin{pmatrix} x_s \\ y_s \end{pmatrix} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}\begin{pmatrix} x_l \\ y_l \end{pmatrix}. \tag{14}$$

Differentiating twice yields the relation between the acceleration of x_s and y_s with respect to that of x_l and y_l, which is given by

TABLE 4 Main acceleration components linking $\ddot{x}_s$ to $\ddot{x}_l$. These terms form the main components required to decouple the stage forces from the lens coordinates to stage coordinates. For the vertical directions, a similar result is obtained.

Component	Meaning
$\ddot{x}_l$	The main element, acceleration in x
$\ddot{y}_l\theta$	A rotated chuck requires a small force proportional to $\ddot{y}_l$, because x_s and x_l are no longer aligned
$2\dot{y}_l\dot{\theta}$	Coriolis force, proportional to the product of these two velocities
$y_l\ddot{\theta}$	Main crosstalk component, acceleration in θ requires acceleration in x to keep the same point of the wafer under the lens

$$\begin{aligned}\ddot{x}_s = {} & \ddot{x}_l\cos(\theta) - 2\dot{x}_l\dot{\theta}\sin(\theta) - x_l(\ddot{\theta}\sin(\theta) + \dot{\theta}^2\cos(\theta)) \\ & - \ddot{y}_l\sin(\theta) - 2\dot{y}_l\dot{\theta}\cos(\theta) - y_l(\ddot{\theta}\cos(\theta) - \dot{\theta}^2\sin(\theta)),\end{aligned} \tag{15}$$

$$\begin{aligned}\ddot{y}_s = {} & \ddot{x}_l\sin(\theta) + 2\dot{x}_l\dot{\theta}\cos(\theta) + x_l(\ddot{\theta}\cos(\theta) - \dot{\theta}^2\sin(\theta)) \\ & + \ddot{y}_l\cos(\theta) - 2\dot{y}_l\dot{\theta}\sin(\theta) - y_l(\ddot{\theta}\sin(\theta) + \dot{\theta}^2\cos(\theta)).\end{aligned} \tag{16}$$

Using the small angle approximations $\sin(\theta) \approx \theta$, $\cos(\theta) \approx 1$ in (15) and (16) yields

$$\ddot{x}_s \approx \ddot{x}_l - 2\dot{x}_l\dot{\theta}\theta - x_l(\ddot{\theta}\theta + \dot{\theta}^2) - \ddot{y}_l\theta - 2\dot{y}_l\dot{\theta} - y_l(\ddot{\theta} - \dot{\theta}^2\theta), \tag{17}$$

$$\ddot{y}_s \approx \ddot{x}_l\theta + 2\dot{x}_l\dot{\theta} + x_l(\ddot{\theta} - \dot{\theta}^2\theta) + \ddot{y}_l - 2\dot{y}_l\dot{\theta}\theta - y_l(\ddot{\theta}\theta + \dot{\theta}^2). \tag{18}$$

Furthermore, assuming $\dot{\theta}\theta \approx 0$, $\ddot{\theta}\theta \approx 0$, $\dot{\theta}^2 \approx 0$, (17) and (18) become

$$\ddot{x}_s \approx \ddot{x}_l - \ddot{y}_l\theta - 2\dot{y}_l\dot{\theta} - y_l\ddot{\theta}, \tag{19}$$

$$\ddot{y}_s \approx \ddot{y}_l + \ddot{x}_l\theta + 2\dot{x}_l\dot{\theta} + x_l\ddot{\theta}. \tag{20}$$

During an exposure scan, θ is kept constant as much as possible, which makes the assumption $\dot{\theta}^2 \approx 0$ valid. Rotation velocities $\dot{\chi}$ and $\dot{\psi}$ can be larger than $\dot{\theta}$ due to wafer surface tracking. We will nevertheless also assume $\dot{\chi}^2 \approx 0$ and $\dot{\psi}^2 \approx 0$ because the allowed position errors in χ and ψ, which contribute to focus errors, are much larger. Table 4 lists the main components in the formula for $\ddot{x}_s$. Apart from the main x acceleration component, $\ddot{x}_l$, the term $y_l\ddot{\theta}$ plays the largest role. This term dictates that the stage center needs an extra x acceleration proportional to the θ acceleration, multiplied by the y distance of the stage center to the lens center. Without this term, an acceleration of θ would lead to a shift in the x direction of the wafer with respect to the lens, depending on the y position. Including this term ensures that the expose position on the wafer remains under the lens center in the presence of an acceleration in θ. The stage now rotates around the lens center instead of its center of mass. Especially in the vertical directions, the applicable rotations may show a high acceleration, depending on the vertical topology of the wafer surface.

The presence of the Coriolis term $2\dot{y}_l\dot{\theta}$ is explained in Figure 28 for the y and χ directions. From left to right, the stage is moving with constant velocity in the $+y$ direction and simultaneously rotating in the $+\chi$ direction. The defocus z at the point of interest increases quadratically with time. A constant z force proportional to the product of $\dot{y}$ and $\dot{\chi}$ compensates for this effect and keeps the wafer in the focal plane of the lens.

From (19) and (20), the x and y force in stage coordinates can be calculated from the respective forces in lens coordinates by multiplication by the stage mass m and inertia J_z around the z axis, that is,

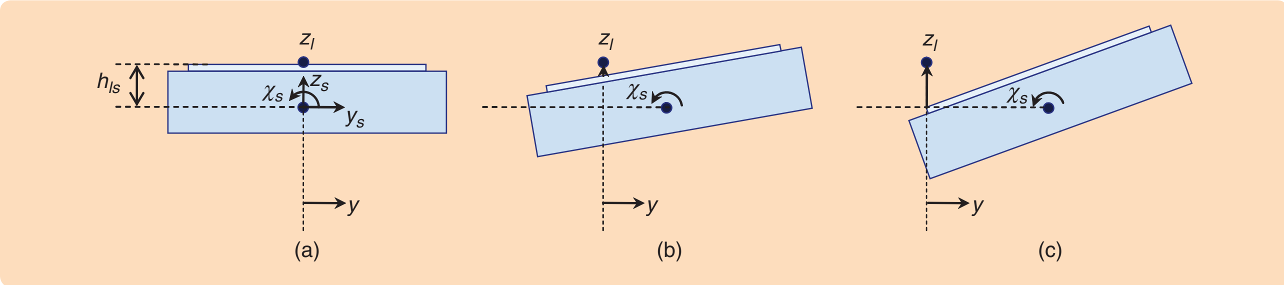

FIGURE 28 Coriolis force in the z direction. A constant stage velocity in the $+y$ direction, in combination with a constant velocity in the $+\chi$ direction, results in a quadratic z error with time. A constant force F_z, proportional to the product of both velocities, compensates for this effect.

$$F_{xs} = F_{xl} - F_{yl}\theta - 2m\dot{y}_l\dot{\theta} - y_l\frac{m}{J_z}T_z, \tag{21}$$

$$F_{ys} = F_{yl} + F_{xl}\theta + 2m\dot{x}_l\dot{\theta} + x_l\frac{m}{J_z}T_z. \tag{22}$$

The controller-generated torque T_z around the z axis results in an acceleration in θ direction. Calculating F_{xs} and F_{ys} requires variables that are not directly available from measurements, such as $\dot{y}_l$ and $\dot{\theta}$. Furthermore, although the measured variables y_l and θ are available, it is desirable to avoid using them directly in the compensation matrix because a feedback loop is created that may endanger stability. Therefore, the setpoints of these variables are used, which is allowed because the difference between the actual position and velocity and the respective setpoints is much smaller than the absolute values of the setpoints.

Figure 29 shows a block scheme of the decoupling presented in (21) and (22). The inputs are formed by setpoint values and horizontal control forces in lens coordinates F_{xl}, F_{yl}, and T_{zl}. The gain scheduling output control forces and torques are in the stage's coordinate system.

Vertical Gain Scheduling

In the vertical direction a similar coupling is present, with the additional element that the stage's center of mass is located beneath the wafer surface by a distance h_{ls}, as depicted in Figure 19. A rotation about the x or y axis then leads to a shift in both the z and either the y or x positions, respectively. Hence, a torque T_x needs to be accompanied by forces F_z and F_y to keep the wafer surface in the lens' focal plane and to prevent an image shift in y. Similar calculations as in the horizontal case lead to the compensating forces

$$F_{xs} = F_{xl} + F_{zl}\psi + 2m\dot{z}_l\dot{\psi} + \frac{m}{J_y}(z_l - h_{ls})T_y, \tag{23}$$

$$F_{zs} = F_{zl} - F_{xl}\psi + F_{yl}\chi - 2m\dot{x}_l\dot{\psi} + 2m\dot{y}_l\dot{\chi} + \frac{m}{J_y}x_lT_y - \frac{m}{J_x}y_lT_x, \tag{24}$$

$$F_{ys} = F_{yl} - F_{zl}\chi - 2m\dot{z}_l\dot{\chi} - \frac{m}{J_x}(z_l - h_{ls})T_x. \tag{25}$$

The total horizontal forces F_{xs} and F_{ys} are formed by combining (21), (22) with (23)–(25), taking into account that F_{xl} and F_{yl} need to be included only once. The torques T_x, T_y, and T_z do not need any adjustment and hence are the same in the lens and stage coordinate systems.

Figure 30 shows the 6×6 Bode plot of the stage's transfer function, assuming the stage itself is completely rigid, without internal dynamics. On the diagonal, the mass lines are visible. In all locations in the (x, y) plane, crosstalk is visible from T_x to y and from T_y to x, depending on the height h_{ls}. If the stage moves in the $+x$ direction, additional cross terms occur from T_z to y and T_y to z. Similarly, a movement in $+y$ direction result in cross terms from T_z to x and T_x to z. When applying the gain scheduling technique as described above, only the diagonal terms remain and decoupling is exact.

Effect of Stage Dynamics

As in the stepper and scanner cases described above, the motor is not rigidly connected to the mirror block. Thus far, we have considered only the 1DOF case, but to investigate the effects of the limited stiffness on

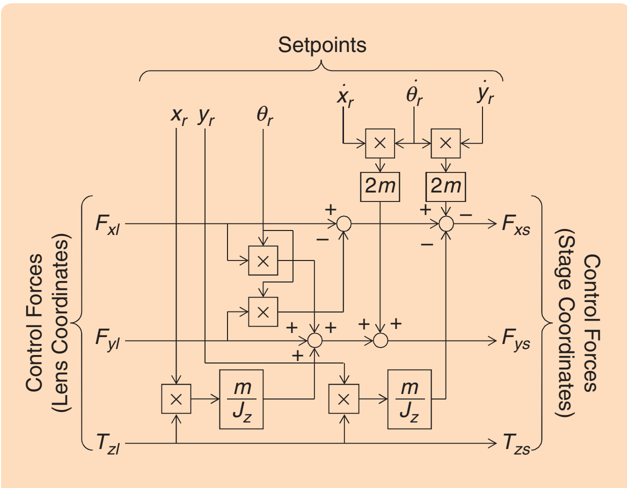

FIGURE 29 Gain scheduling of the horizontal control forces. Forces in the x and y directions are corrected by additional forces proportional to the torque T_z. In addition, the scheme shows a Coriolis correction as well as a correction for a nominal value of θ not equaling zero.

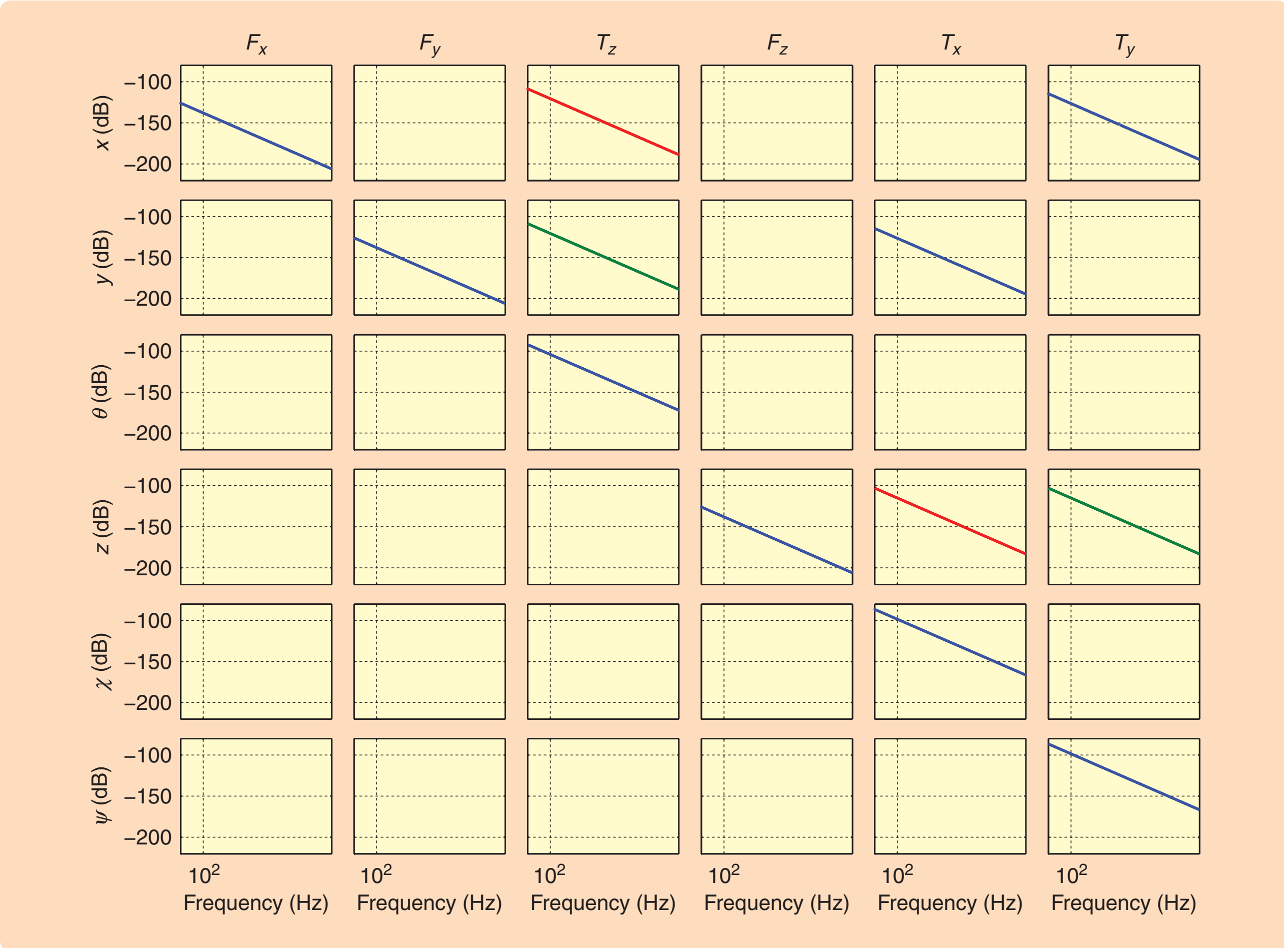

FIGURE 30 Stage transfer function. Each column represents the responses in all degrees of freedom to a force or torque in the direction indicated above the column. The response depends on the position of the stage in the horizontal plane. The blue lines show the response at $(x, y) = (0, 0)$, the green lines at $(x, y) = (0.15, 0)$, and the red lines at $(x, y) = (0, 0.15)$. All off-diagonal terms vanish when applying gain scheduling, leaving a perfectly decoupled 6×6 system allowing six single-input, single-output controllers to be used.

decoupling quality, and hence control behavior, we need a 6DOF analysis. Figure 26 shows a basic picture of a motor block providing the stage forces, connected to the mirror block by means of a limited stiffness. In the x and y directions, the motor block is coupled with nominal 800- and 900-Hz resonances, respectively, while in the z direction a frequency of 500 Hz is present. Torsional modes have higher frequencies. The controller bandwidth in the horizontal directions is increased to 200 Hz, while the vertical directions use a 100-Hz bandwidth, dictated by the lower frequency of the first resonance. When applying the rigid-body gain scheduling technique, the decoupling result of Figure 31 occurs. The blue curves show the 6×6 transfer function without gain scheduling, while the red curves show the transfer function with the original gain scheduling, at the position $(x, y) = (150, 150)$ mm. Two main effects are observed.

First, cross terms are no longer completely reduced to zero. In some directions, the original transfer function of an applied force or torque to the resulting position shows a mass behavior having a slope of -40 dB/decade. Instead of completely canceling these cross terms, a stiffness behavior arises, which has a flat amplitude line for lower frequencies. Second, some new cross terms are now present, which are absent without gain scheduling. Due to the motor connection to the stage, T_y creates extra cross terms.

Nonperfect decoupling implies that each controller may experience disturbances from other motion directions, which the controller needs to tackle with sufficient disturbance rejection to reduce the error to within specification. In addition, note that the stage dynamic behavior is built up from its individual dynamic modes, which are weighted differently depending on the stage's position. For example, two modes could be a z resonance and a χ resonance of the mirror block with respect to the motor. If $y = 0$, the measured z position consists of mainly the z mode. For positive and negative y positions, the contribution of the χ mode to the measured z position has the

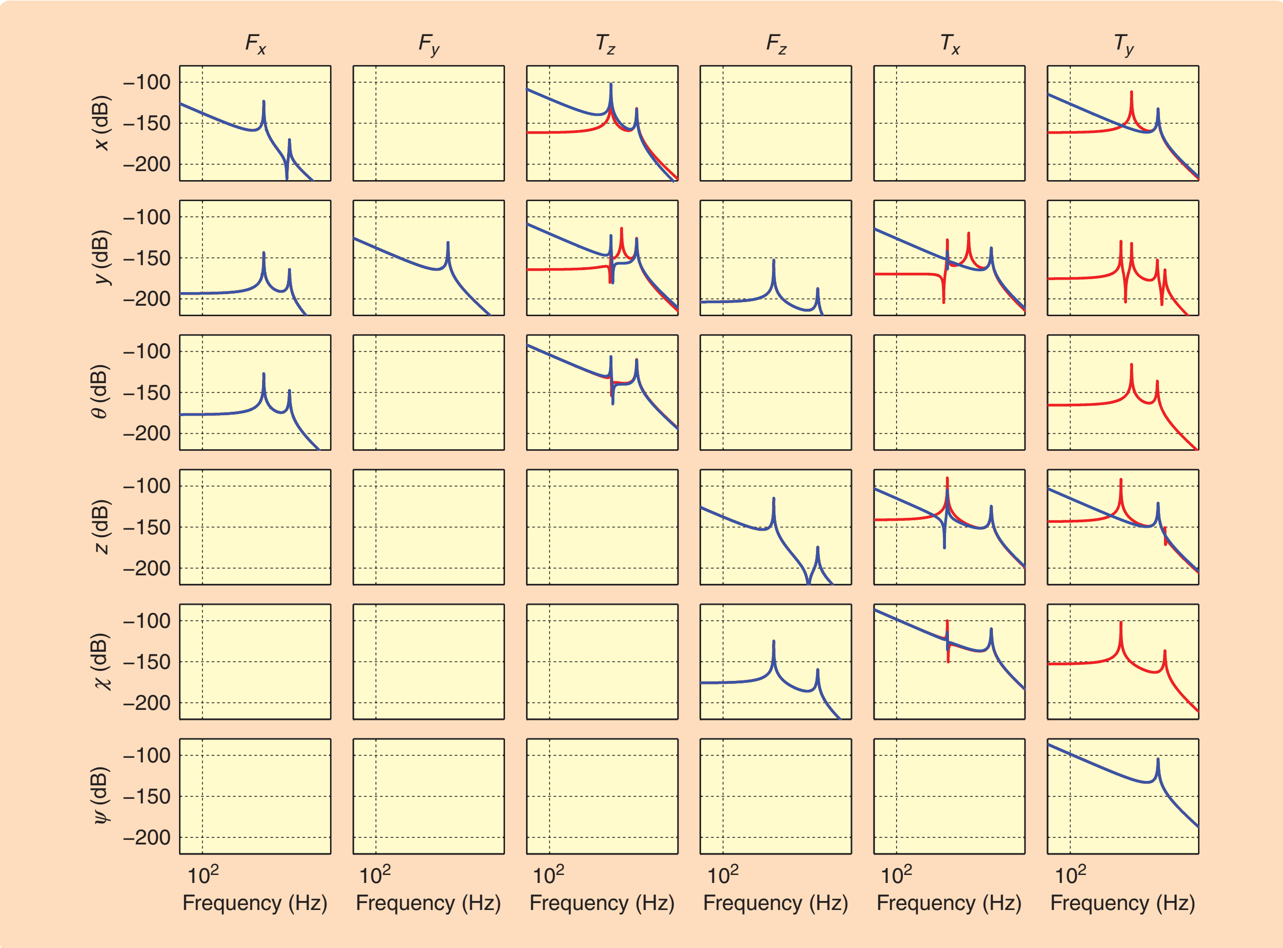

FIGURE 31 Stage transfer function with dynamics. The blue lines show the response without gain scheduling at $(x, y) = (0.15, 0.15)$; the red lines show the decoupled system. Gain scheduling is intended for the rigid-body system and thus does not perfectly decouple the system with flexible dynamics. For low frequencies, all mass lines are converted into stiffness lines with reduced coupling. Some additional cross terms arise, which are not present without decoupling. When using six single-input, single-output controllers, nonperfect decoupling results in disturbance forces from other motion directions, which need to be addressed by disturbance rejection of the controller, requiring sufficient controller bandwidth.

opposite sign. As illustration, figures 32 and 33 show the Bode and Nyquist plots of the z-controller open loop for three y positions. It can be seen that the resonance frequencies corresponding to the modes stay the same, but the antiresonance frequency shifts, even leading to a 180° phase shift at the 1.8-kHz χ resonance. The controller needs to be able to deal with all possible transfer functions, which in this case is not a problem because of the low amplitude of the 1.8-kHz mode. Figure 34 shows a time-domain result in which both the vibrations excited by the setpoint, as well as crosstalk from vertical directions during the scan, are visible. In this particular example, MA and MSD values are in the single-nanometer range, despite the nonexact decoupling.

Other Critical Control Aspects

In addition to decoupling and controller design itself, several other aspects form a key factor in achieving accurate stage positioning. Some of these aspects are mentioned here.

Disturbance Rejection

Rejection of measurement system disturbances, such as lens vibrations showing up in the interferometer position measurement, and input disturbances, such as noise induced by immersion water flow or amplifier noise, is also required. Reducing these disturbance sources is the first step, after which specific controller design comes in. Since many disturbance sources have a specific spectral content, frequency-domain controller design is favored.

Setpoint Design

The stage's setpoint needs to be synchronized in all directions to enable wafer stage and reticle stage to scan synchronously and thus minimize the position error of the image on the wafer. The relative timing of reticle

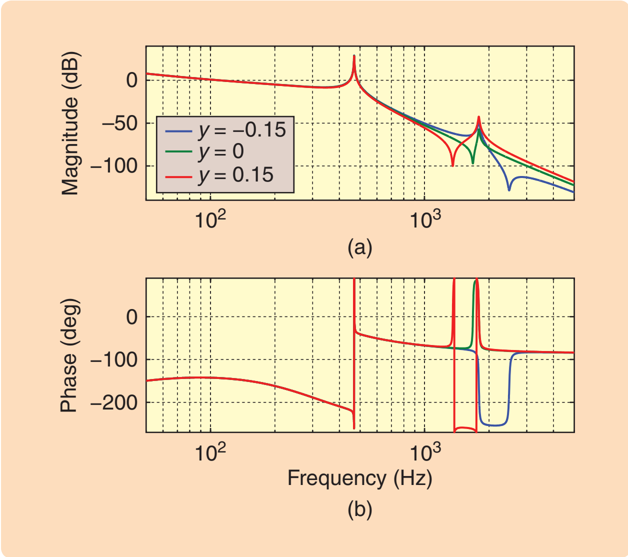

FIGURE 32 Bode plot of the z-controller open loop for three y positions. The resonances in all three cases coincide with the 500-Hz z mode and 1800-Hz χ mode. Because the χ mode is observed differently in the z measurement for different y positions, the antiresonance frequency and phase behavior in the transfer function change. This change is not a problem because the magnitude of the 1800-Hz resonance is small.

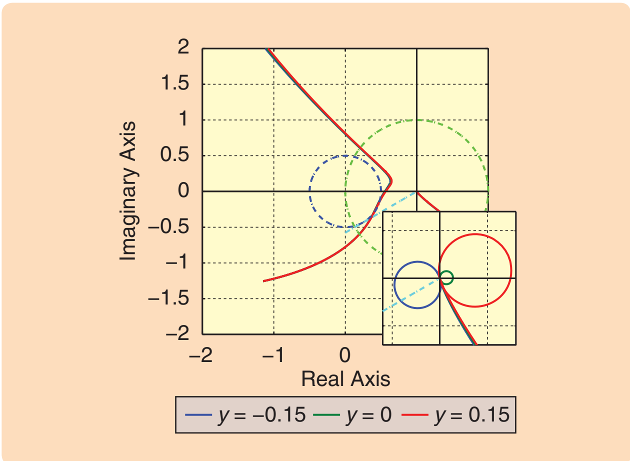

FIGURE 33 Nyquist plot of the z-controller open loop for three y positions. The inset shows the magnified behavior around the origin. This graph further elucidates the position dependence of the dynamical behavior.

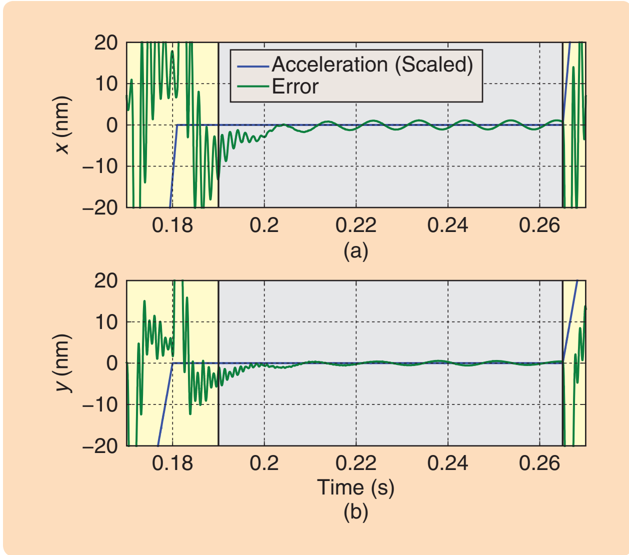

FIGURE 34 x and y time response of the wafer stage with horizontal dynamics. The settling behavior related to the horizontal dynamics caused by the motor-to-mirror block connection is visible. A more advanced feedforward can reduce this error. The sinusoidal behavior during the scan is caused by crosstalk from the vertical directions, which are controlled to track the wafer surface. This crosstalk is the result of nonideal decoupling due to the presence of the nonrigid motor-to-mirror block connection.

stage and wafer stage setpoints is essential since at a scan speed of 1 m/s and a projection optics length of 1 m, an image position error of 3 nm is induced by the speed of light. This error is unacceptable in terms of the required positioning accuracy. In addition, the setpoint order, defined by the number of continuous derivatives of the position reference, determines its frequency content, which influences the stage response. Specific setpoint design can help avoid excitation of specific resonances in the system.

Setpoint Feedforward

Thus far we have used only acceleration feedforward by multiplying the commanded acceleration by the stage mass to create a feedforward force. For a stage mass of 20 kg and an acceleration of 30 m/s^2, 600 N is needed to accelerate the stage. A mismatch in the feedforward gain results in a controller error. With a 200-Hz bandwidth, 1-nm error remains if the feedforward has an accuracy of 99.99%. Therefore, at most a 60-mN mismatch in feedforward force is allowed on the total 600 N for a 1-nm error. A mismatch can be caused by a gain deviation in the path from calculated control force up to the actual stage acceleration, such as amplifier gain, motor constant, and wafer mass.

Not only the magnitude, but also correct timing in applying the feedforward signal, with respect to the position reference, is essential. Compensation of the feedforward timing with respect to the position-reference timing is required because of the presence of time delay in the system. This time delay is mostly caused by digital-to-analog conversion and calculation time of the controller algorithm. An error of 1 nm remains when the feedforward timing is off by 3 μs in a 10-kHz sampled system. Feedforward magnitude and timing are automatically calibrated for each stage.

In addition to the setpoint acceleration feedforward, extra terms related to higher derivatives of the position setpoint can be used. *Jerk, snap, crackle,* and *pop* refer to the third, fourth, fifth, and sixth derivatives of the position reference [9]. Using these higher derivatives in the feedforward path allows construction of a better plant inverse in the feedforward. This way, the feedforward takes care of the stage dynamics, resulting in a smaller controller error for the feedback controller to deal with.

Other Feedforwards

The stage needs to track the projection lens, which it does according to its sensitivity function because the stage position is measured with respect to the projection lens. If acceleration data of the projection lens is available, such a measurement can be used to create an additional feedforward signal. Additional feedforward strategies include position-dependent disturbance force compensation, stage-to-stage feedforward, and long-stroke-to-short-stroke feedforward to compensate parasitic stiffness and damping between these two.

Calibrations

Although all stages are designed to be equal, some production tolerances lead to variation in, for example, actuator driving force directions, motor constants, and dynamic behavior. Individual automatic calibration of these variables may be needed to create a system that conforms to its specification. In particular, the gain balancing and gain scheduling parameters, motor gain and linearity, interferometer beam pointing, mirror surface unflatness, and timing differences between stage controllers, are all automatically calibrated per machine.

CONCLUSIONS

The optical exposure process used to create ICs requires a positioning accuracy of the wafer and the reticle that is a fraction of the smallest feature being manufactured. This minimum feature size has decreased from 350 nm in 1994 to 20 nm today. At the same time, the wafer surface area has doubled, and the number of wafers produced per hour has tripled.

The lithographic process requires mechanical positioning of the wafer and reticle during exposure of each IC. Improvements in speed, acceleration, and position accuracy of the stages have enabled the progress in machine throughput and pattern shrink. Stage architecture and machine layout have created a cleaner and more vibration-free environment by reducing the effects of reaction forces and external disturbances. The stage behavior itself, combined with compensation of known nonlinearities, has become virtually linear. Classical multiple SISO control techniques are used, aided by careful feedforward design and actuator coordinate decoupling. Frequency-domain design techniques provide the ability to adjust the controller for specific disturbance frequencies. At the same time, time-domain behavior, related to the application of the setpoint profiles, can be straightforwardly anticipated during the design. Today, accelerations of 20–40 m/s^2, wafer scan speeds of 0.6–1 m/s, a settling time of a few milliseconds, and a position accuracy of a few nanometers, are combined in production-worthy industrial tools.

Since the position-accuracy requirements scale with the critical dimension of the manufactured IC, future systems require a further reduction of the positioning error. At the same time, acceleration and scan speed need to increase further to compensate for the increased cost of new technology. Mechatronic challenges include using planar actuators as long-stroke drive systems and planar encoders for stage position measurement. A next step in critical dimension reduction will be provided by extreme-ultraviolet lithography making use of a 13.5 nm radiation wavelength. This radiation does not pass through any material, including air, requiring all-vacuum stages and all-mirror optics replacing refractive optics. New, more advanced control technology needs to be developed to enable further advances in IC functionality.

AUTHOR INFORMATION

Hans Butler joined ASML in 1991 after obtaining his M.Sc. and Ph.D. in adaptive control at Delft University of Technology in 1986 and 1990, respectively. He has worked on multiple components in lithographic tools as a control engineer, such as handlers, illuminators, projection systems, and stages. Within the Mechatronic Systems Development Department, he has led groups working on actuators, opto-mechatronics, and, more recently, dynamics, vibration isolation, and damping. Currently he is an ASML fellow with dynamics and control as his main focus area.

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